conditional on announcing $\hat{\theta}_i$. Because of the form of agents' utility functions, we can write agent i's expected utility when he is type θ_i and announces his type to be $\hat{\theta}_i$ (assuming that all agents $j \neq i$ tell the truth) as[32]

$$E_{\theta_{-i}}[u_i(f(\hat{\theta}_i, \theta_{-i}), \theta_i) | \theta_i] = \theta_i \bar{v}_i(\hat{\theta}_i) + \bar{t}_i(\hat{\theta}_i). \tag{23.D.10}$$

It is also convenient to define for each i the function

$$U_i(\theta_i) = \theta_i \bar{v}_i(\theta_i) + \bar{t}_i(\theta_i),$$

giving agent i's expected utility from the mechanism conditional on his type being θ_i when he and all other agents report their true types.

Proposition 23.D.2: The social choice function $f(\cdot) = (k(\cdot), t_1(\cdot), \ldots, t_I(\cdot))$ is Bayesian incentive compatible if and only if, for all $i = 1, \ldots, I$,

(i) $\bar{v}_i(\cdot)$ is nondecreasing. (23.D.11)

(ii) $U_i(\theta_i) = U_i(\underline{\theta}_i) + \int_{\underline{\theta}_i}^{\theta_i} \bar{v}_i(s)\, ds$ for all θ_i. (23.D.12)

Proof: (i) *Necessity.* Bayesian incentive compatibility implies that for each $\hat{\theta}_i > \theta_i$ we have

$$U_i(\theta_i) \geq \theta_i \bar{v}_i(\hat{\theta}_i) + \bar{t}_i(\hat{\theta}_i) = U_i(\hat{\theta}_i) + (\theta_i - \hat{\theta}_i)\bar{v}_i(\hat{\theta}_i)$$

and

$$U_i(\hat{\theta}_i) \geq \hat{\theta}_i \bar{v}_i(\theta_i) + \bar{t}_i(\theta_i) = U_i(\theta_i) + (\hat{\theta}_i - \theta_i)\bar{v}_i(\theta_i).$$

Thus,

$$\bar{v}_i(\hat{\theta}_i) \geq \frac{U_i(\hat{\theta}_i) - U_i(\theta_i)}{\hat{\theta}_i - \theta_i} \geq \bar{v}_i(\theta_i). \tag{23.D.13}$$

Expression (23.D.13) immediately implies that $\bar{v}_i(\cdot)$ must be nondecreasing (recall that we have taken $\hat{\theta}_i > \theta_i$). In addition, letting $\hat{\theta}_i \to \theta_i$ in (23.D.13) implies that for all θ_i we have

$$U_i'(\theta_i) = \bar{v}_i(\theta_i)$$

and so

$$U_i(\theta_i) = U_i(\underline{\theta}_i) + \int_{\underline{\theta}_i}^{\theta_i} \bar{v}_i(s)\, ds \qquad \text{for all } \theta_i.$$

(ii) *Sufficiency.* Consider any θ_i and $\hat{\theta}_i$ and suppose without loss of generality that $\theta_i > \hat{\theta}_i$. If (23.D.11) and (23.D.12) hold, then

$$\begin{aligned} U_i(\theta_i) - U_i(\hat{\theta}_i) &= \int_{\hat{\theta}_i}^{\theta_i} \bar{v}_i(s)\, ds \\ &\geq \int_{\hat{\theta}_i}^{\theta_i} \bar{v}_i(\hat{\theta}_i)\, ds \\ &= (\theta_i - \hat{\theta}_i)\bar{v}_i(\hat{\theta}_i). \end{aligned}$$

32. Observe that the agent's preferences here over his expected benefit $\bar{v}_i$ and expected transfer $\bar{t}_i$ satisfy the single-crossing property that played a prominent role in Sections 13.C and 14.C.

Hence,

$$U_i(\theta_i) \geq U_i(\hat{\theta}_i) + (\theta_i - \hat{\theta}_i)\bar{v}_i(\hat{\theta}_i) = \theta_i\bar{v}_i(\hat{\theta}_i) + \bar{t}_i(\hat{\theta}_i).$$

Similarly, we can derive that

$$U_i(\hat{\theta}_i) \geq U_i(\theta_i) + (\hat{\theta}_i - \theta_i)\bar{v}_i(\theta_i) = \hat{\theta}_i\bar{v}_i(\theta_i) + \bar{t}_i(\theta_i).$$

So $f(\cdot)$ is Bayesian incentive compatible. ■

Proposition 23.D.2 shows that to identify all Bayesian incentive compatible social choice functions in the linear setting, we can proceed as follows: First identify which functions $k(\cdot)$ lead every agent i's expected benefit function $\bar{v}_i(\cdot)$ to be nondecreasing. Then, for each such function, identify the expected transfer functions $\bar{t}_1(\cdot), \ldots, \bar{t}_I(\cdot)$ that satisfy condition (23.D.12) of the proposition. Substituting for $U_i(\cdot)$, these are precisely the expected transfer functions that satisfy, for $i = 1, \ldots, I$,

$$\bar{t}_i(\theta_i) = \bar{t}_i(\underline{\theta}_i) + \underline{\theta}_i v_i(\underline{\theta}_i) - \theta_i v_i(\theta_i) + \int_{\underline{\theta}_i}^{\theta_i} \bar{v}_i(s)\, ds$$

for some constant $\bar{t}_i(\underline{\theta}_i)$. Finally, choose any set of transfer functions $(t_1(\theta), \ldots, t_I(\theta))$ such that $E_{\theta_{-i}}[t_i(\theta_i, \theta_{-i})] = \bar{t}_i(\theta_i)$ for all θ_i. In general, there are many such functions $t_i(\cdot, \cdot)$; one, for example, is simply $t_i(\theta_i, \theta_{-i}) = \bar{t}_i(\theta_i)$.[33]

We now illustrate one implication of this characterization result for the auction setting introduced in Example 23.B.4. Some further implications of Proposition 23.D.2 are derived in Sections 23.E and 23.F.

Auctions: the revenue equivalence theorem

Let us consider again the auction setting introduced in Example 23.B.4: Agent 0 is the seller of an indivisible object from which he derives no value, and agents $1, \ldots, I$ are potential buyers.[34] It will be convenient, however, to generalize the set of possible alternatives relative to those considered in Example 23.B.4 by allowing for a *random* assignment of the object. Thus, we now take $y_i(\theta)$ to be buyer i's *probability* of getting the object when the vector of announced types is $\theta = (\theta_1, \ldots, \theta_I)$. Buyer i's expected utility when the profile of types for the I buyers is $\theta = (\theta_1, \ldots, \theta_I)$ is then $\theta_i y_i(\theta) + t_i(\theta)$. Note that buyer i is risk neutral with respect to lotteries both over transfers and over the allocation of the good.

This setting corresponds in the framework studied in Proposition 23.D.2 to the case where we take $k = (y_1, \ldots, y_I)$, $K = \{(y_1, \ldots, y_I)\colon y_i \in [0, 1]$ for all $i = 1, \ldots, I$ and $\sum_i y_i \leq 1\}$, and $v_i(k) = y_i$. Thus, to apply Proposition 23.D.2 we can write $\bar{v}_i(\hat{\theta}_i) = \bar{y}_i(\hat{\theta}_i)$, where $\bar{y}_i(\hat{\theta}_i) = E_{\theta_{-i}}[y_i(\hat{\theta}_i, \theta_{-i})]$ is the probability that i gets the object conditional on announcing his type to be $\hat{\theta}_i$ when agents $j \neq i$ announce their types truthfully, and $U_i(\theta_i) = \theta_i \bar{y}_i(\theta_i) + \bar{t}_i(\theta_i)$.

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33. However, if we wish the social choice function $f(\cdot) = (k(\cdot), t_1(\cdot), \ldots, t_I(\cdot))$ to satisfy some further properties, such as budget balance, only a subset (possibly an empty one) of the transfer functions generating the expected transfer functions $(\bar{t}_1(\theta_1), \ldots, \bar{t}_I(\theta_I))$ may have these properties.

34. We note that our assumption that the seller in an auction setting derives no value from the object is not necessary for the revenue equivalence theorem. (As we shall see, the result characterizes the expected revenues generated for the seller in different auctions, and so is valid for any utility function that the seller might have.) In the absence of this assumption, however, the seller in an auction will generally care about more than just the expected revenue he receives.

We can now establish a remarkable result, known as the *revenue equivalence theorem*.[35]

Proposition 23.D.3: (*The Revenue Equivalence Theorem*) Consider an auction setting with I risk-neutral buyers, in which buyer i's valuation is drawn from an interval $[\underline{\theta}_i, \bar{\theta}_i]$ with $\underline{\theta}_i \neq \bar{\theta}_i$ and a strictly positive density $\phi_i(\cdot) > 0$, and in which buyers' types are statistically independent. Suppose that a given pair of Bayesian Nash equilibria of two different auction procedures are such that for every buyer i: (i) For each possible realization of $(\theta_1, \ldots, \theta_I)$, buyer i has an identical probability of getting the good in the two auctions; and (ii) Buyer i has the same expected utility level in the two auctions when his valuation for the object is at its lowest possible level. Then these equilibria of the two auctions generate the same expected revenue for the seller.

Proof: By the revelation principle, we know that the social choice function that is (indirectly) implemented by the equilibrium of any auction procedure must be Bayesian incentive compatible. Thus, we can establish the result by showing that if two Bayesian incentive compatible social choice functions in this auction setting have the same functions $(y_1(\theta), \ldots, y_I(\theta))$ and the same values of $(U_1(\underline{\theta}_1), \ldots, U_I(\underline{\theta}_I))$ then they generate the same expected revenue for the seller.

To show this, we derive an expression for the seller's expected revenue from an arbitrary Bayesian incentive compatible mechanism. Note, first, that the seller's expected revenue is equal to $\sum_{i=1}^{I} E[-t_i(\theta)]$. Now,

$$\begin{aligned}
E[-t_i(\theta)] &= E_{\theta_i}[-\bar{t}_i(\theta_i)] \\
&= \int_{\underline{\theta}_i}^{\bar{\theta}_i} [\bar{y}_i(\theta_i)\theta_i - U_i(\theta_i)]\phi_i(\theta_i)\, d\theta_i \\
&= \int_{\underline{\theta}_i}^{\bar{\theta}_i} \left(\bar{y}_i(\theta_i)\theta_i - U_i(\underline{\theta}_i) - \int_{\underline{\theta}_i}^{\theta_i} \bar{y}_i(s)\, ds\right)\phi_i(\theta_i)\, d\theta_i \\
&= \left[\int_{\underline{\theta}_i}^{\bar{\theta}_i} \left(\bar{y}_i(\theta_i)\theta_i - \int_{\underline{\theta}_i}^{\theta_i} \bar{y}_i(s)\, ds\right)\phi_i(\theta_i)\, d\theta_i\right] - U_i(\underline{\theta}_i).
\end{aligned}$$

Moreover, integration by parts implies that

$$\begin{aligned}
\int_{\underline{\theta}_i}^{\bar{\theta}_i} \left(\int_{\underline{\theta}_i}^{\theta_i} \bar{y}_i(s)\, ds\right)\phi_i(\theta_i)\, d\theta_i &= \left(\int_{\underline{\theta}_i}^{\bar{\theta}_i} \bar{y}_i(\theta_i)\, d\theta_i\right) - \left(\int_{\underline{\theta}_i}^{\bar{\theta}_i} \bar{y}_i(\theta_i)\Phi_i(\theta_i)\, d\theta_i\right) \\
&= \int_{\underline{\theta}_i}^{\bar{\theta}_i} \bar{y}_i(\theta_i)(1 - \Phi_i(\theta_i))\, d\theta_i.
\end{aligned}$$

Substituting, we see that

$$E[-\bar{t}_i(\theta_i)] = \left[\int_{\underline{\theta}_i}^{\bar{\theta}_i} \bar{y}_i(\theta_i)\left(\theta_i - \frac{1 - \Phi_i(\theta_i)}{\phi_i(\theta_i)}\right)\phi_i(\theta_i)\, d\theta_i\right] - U_i(\underline{\theta}_i), \qquad (23.D.14)$$

35. Versions of the revenue equivalence theorem have been derived by many authors; see McAfee and McMillan (1987) and Milgrom (1987) for references as well as for a further discussion of the result.

or, equivalently,

$$E[-\bar{t}_i(\theta_i)] =$$

$$\left[\int_{\underline{\theta}_1}^{\bar{\theta}_1} \cdots \int_{\underline{\theta}_I}^{\bar{\theta}_I} y_i(\theta_1, \ldots, \theta_I)\left(\theta_i - \frac{1 - \Phi_i(\theta_i)}{\phi_i(\theta_i)}\right)\left(\prod_{j=1}^{I} \phi_j(\theta_j)\right) d\theta_I \cdots d\theta_1\right] - U_i(\underline{\theta}_i). \tag{23.D.15}$$

Thus, the seller's expected revenue is equal to

$$\left[\int_{\underline{\theta}_1}^{\bar{\theta}_1} \cdots \int_{\underline{\theta}_I}^{\bar{\theta}_I} \left[\sum_{i=1}^{I} y_i(\theta_1, \ldots, \theta_I)\left(\theta_i - \frac{1 - \Phi_i(\theta_i)}{\phi_i(\theta_i)}\right)\right]\left(\prod_{j=1}^{I} \phi_j(\theta_j)\right) d\theta_I \cdots d\theta_1\right] - \sum_{i=1}^{I} U_i(\underline{\theta}_i). \tag{23.D.16}$$

By inspection of (23.D.16), we see that any two Bayesian incentive compatible social choice functions that generate the same functions $(y_1(\theta), \ldots, y_I(\theta))$ and the same values of $(U_1(\underline{\theta}_1), \ldots, U_I(\underline{\theta}_I))$ generate the same expected revenue for the seller. ∎

As an example of the application of Proposition 23.D.3, consider the equilibria of the first-price and second-price sealed-bid auctions that we identified in Examples 23.B.5 and 23.B.6 (where the buyers' valuations were independently drawn from the uniform distribution on [0, 1]). For these equilibria, the conditions of the revenue equivalence theorem are satisfied: in both auctions the buyer with the highest valuation always gets the good and a buyer with a zero valuation has an expected utility of zero. Thus, the revenue equivalence theorem tells us that the seller receives exactly the same level of expected revenue in these equilibria of the two auctions (you can confirm this fact in Exercise 23.D.3). More generally, it can be shown that in any *symmetric auction setting* (i.e., one where the buyers' valuations are independently drawn from identical distributions), the conditions of the revenue equivalence theorem will be met for *any* Bayesian Nash equilibrium of the first-price sealed-bid auction and the (dominant strategy) equilibrium of the second-price sealed-bid auction (see Exercise 23.D.4 for a consideration of symmetric equilibria in these settings). We can conclude from Proposition 23.D.3, therefore, that in any such setting the first-price and second-price sealed-bid auctions generate exactly the same revenue for the seller.

23.E Participation Constraints

In Sections 23.B to 23.D, we have studied the constraints that the presence of private information puts on the set of implementable social choice functions. Our analysis up to this point, however, has assumed implicitly that each agent i has no choice but to participate in any mechanism chosen by the mechanism designer. That is, agent i's discretion was limited to choosing his optimal actions within those allowed by the mechanism.

In many applications, however, agents' participation in the mechanism is *voluntary*. As a result, the social choice function that is to be implemented by a mechanism must not only be incentive compatible but must also satisfy certain *participation* (or *individual rationality*) *constraints* if it is to be successfully implemented. In this section, we provide a brief discussion of these additional constraints on the set of implementable social choice functions. By way of motivating our study, Example 23.E.1 provides a simple illustration of how the presence of participation constraints may limit the set of social choice functions that can be successfully implemented.

Example 23.E.1: *Participation Constraints in Public Project Choice.* Consider the following simple example of public project choice (recall our initial discussion of public project choice in Example 23.B.3). A decision must be made whether to do a given project or not, so that $K = \{0, 1\}$. There are two agents, 1 and 2. For each agent i, $\Theta_i = \{\underline{\theta}, \bar{\theta}\}$, so that each agent either has a valuation of $\underline{\theta}$, or a valuation of $\bar{\theta}$. We shall assume that $\bar{\theta} > 2\underline{\theta} > 0$. The cost of the project is $c \in (2\underline{\theta}, \bar{\theta})$. Suppose that we want to implement a social choice function having an ex post efficient project choice; that is, one that has $k^*(\theta_1, \theta_2) = 1$ if either θ_1 or θ_2 is equal to $\bar{\theta}$, and $k^*(\theta_1, \theta_2) = 0$ if $\theta_1 = \theta_2 = \underline{\theta}$. In the absence of the need to insure voluntary participation, we know from Section 23.C that we can implement some such social choice function in dominant strategies using a Groves scheme.

Suppose, however, that each agent has the option of withdrawing from the mechanism at any time (perhaps by withdrawing from the group), and that, if he does, he will not enjoy the benefits of the project if it is done, but will also avoid paying any monetary transfers. Can we implement a social choice function that achieves voluntary participation and that has an ex post efficient project choice?[36] The answer is "no." To see this, note that if agent 1 can withdraw at any time, then to insure his participation it must be that $t_1(\underline{\theta}, \bar{\theta}) \geq -\underline{\theta}$. That is, it must be that whenever his valuation for the project is $\underline{\theta}$, he pays no more than $\underline{\theta}$ toward the cost of the project. Now consider what agent 1's transfer must be when both agents announce that they have valuation $\bar{\theta}$: If truth telling is to be a dominant strategy, then $t_1(\bar{\theta}, \bar{\theta})$ must satisfy

$$\bar{\theta}k^*(\bar{\theta}, \bar{\theta}) + t_1(\bar{\theta}, \bar{\theta}) \geq \bar{\theta}k^*(\underline{\theta}, \bar{\theta}) + t_1(\underline{\theta}, \bar{\theta}),$$

or, substituting for $k^*(\bar{\theta}, \bar{\theta})$ and $k^*(\underline{\theta}, \bar{\theta})$,

$$\bar{\theta} + t_1(\bar{\theta}, \bar{\theta}) \geq \bar{\theta} + t_1(\underline{\theta}, \bar{\theta}).$$

Since $t_1(\underline{\theta}, \bar{\theta}) \geq -\underline{\theta}$, this implies that $t_1(\bar{\theta}, \bar{\theta}) \geq -\underline{\theta}$. Thus, we conclude that agent 1 must not make a contribution toward the cost of the project that exceeds $\underline{\theta}$ when $(\theta_1, \theta_2) = (\bar{\theta}, \bar{\theta})$. Moreover, by symmetry, we have exactly the same constraint for agent 2's transfer when $(\theta_1, \theta_2) = (\bar{\theta}, \bar{\theta})$, namely, $t_2(\bar{\theta}, \bar{\theta}) \geq -\underline{\theta}$. Hence, $t_1(\bar{\theta}, \bar{\theta}) + t_2(\bar{\theta}, \bar{\theta}) \geq -2\underline{\theta}$. But if this is so, then because $2\underline{\theta} < c$, the feasibility condition $t_1(\bar{\theta}, \bar{\theta}) + t_2(\bar{\theta}, \bar{\theta}) \leq -c$ cannot be satisfied. We conclude, therefore, that it is impossible to implement a social choice function with an ex post efficient project choice when the agents can withdraw from the mechanism at any time.

Note also that the presence of an "outside agent" (say "agent 0") who does not care about the project decision does not help at all here when that agent can also withdraw from the mechanism at any time. This is because, to insure this agent's participation, his transfer $t_0(\theta_1, \theta_2)$ must be nonnegative for every realization of

36. Note that any social choice function that *fails* to have both agents participate is necessarily *ex post* inefficient because one of the agents is excluded from the benefits of the project.

(θ_1, θ_2). In particular, we must have $t_0(\bar{\theta}, \bar{\theta}) \geq 0$, and so we must fail to satisfy the feasibility condition $t_0(\bar{\theta}, \bar{\theta}) + t_1(\bar{\theta}, \bar{\theta}) + t_2(\bar{\theta}, \bar{\theta}) \leq -c$. ■

As a general matter, we can distinguish among three stages at which participation constraints may be relevant in any particular application. First, as in Example 23.E.1, an agent i may be able to withdraw from the mechanism at the *ex post stage* that arises after the agents have announced their types and an outcome in X has been chosen. Formally, suppose that agent i can receive a utility of $\bar{u}_i(\theta_i)$ by withdrawing from the mechanism when his type is θ_i.[37] Then, to insure agent i's participation, we must satisfy the *ex post participation* (or *individual rationality*) *constraints*[38]

$$u_i(f(\theta_i, \theta_{-i}), \theta_i) \geq \bar{u}_i(\theta_i) \qquad \text{for all } (\theta_i, \theta_{-i}). \tag{23.E.1}$$

In other circumstances, agent i may only be able to withdraw from the mechanism at the *interim stage* that arises after the agents have each learned their type but before they have chosen their actions in the mechanism. Letting $U_i(\theta_i | f) = E_{\theta_{-i}}[u_i(f(\theta_i, \theta_{-i}), \theta_i) | \theta_i]$ denote agent i's *interim expected utility* from social choice function $f(\cdot)$ when his type is θ_i, agent i will participate in a mechanism that implements social choice function $f(\cdot)$ when he is of type θ_i if and only if $U_i(\theta_i | f)$ is not less than $\bar{u}_i(\theta_i)$. Thus, *interim participation* (or *individual rationality*) *constraints* for agent i require that

$$U_i(\theta_i | f) = E_{\theta_{-i}}[u_i(f(\theta_i, \theta_{-i}), \theta_i) | \theta_i] \geq \bar{u}_i(\theta_i) \qquad \text{for all } \theta_i. \tag{23.E.2}$$

In still other cases, agent i might only be able to refuse to participate at the *ex ante stage* that arises before the agents learn their types. Letting $U_i(f) = E_{\theta_i}[U_i(\theta_i | f)] = E[u_i(f(\theta_i, \theta_{-i}), \theta_i)]$ denote agent i's *ex ante expected utility* from a mechanism that implements social choice function $f(\cdot)$, the *ex ante participation* (or *individual rationality*) *constraint* for agent i is

$$U_i(f) \geq E_{\theta_i}[\bar{u}_i(\theta_i)]. \tag{23.E.3}$$

Participation constraints are of the ex ante variety when the agents can agree to be bound by the mechanism prior to learning their types. When, instead, agents know their types prior to the time at which they can agree to be bound by the mechanism, we face interim participation constraints.[39] Finally, if there is no way to bind the

37. We assume that agent i's utility from withdrawal depends only on his own type.

38. We assume throughout that it is always optimal to insure that each agent is always willing to participate. In fact, however, there is no loss of generality from assuming this: When agents can "not participate," any outcome that can arise when some subset I' of the I agents does not participate, say x', should be included in the set X. Because we can always have the mechanism select x' in the circumstances when this subset of agents would have refused to participate, if the set X is defined appropriately we can always replicate the outcome of any mechanism that causes nonparticipation with a mechanism in which all agents are always willing to participate.

39. Recall that the assumption in a Bayesian game that types are drawn from a common prior density is often merely a modeling device for how agents form beliefs about each others' types (see Section 8.E). That is, for analytical purposes we may be representing a setting in which agents' types are already determined but are only privately observed by assuming that there has been a prior random draw of types from a commonly known distribution; but there may not actually be any such prior stage at which the agents could possibly interact.

agents to the assigned outcomes of the mechanism against their will, then we face ex post participation constraints.[40]

Note that if $f(\cdot)$ satisfies (23.E.1), then it satisfies (23.E.2); and, in turn, if it satisfies (23.E.2), then it satisfies (23.E.3). However, the reverse is not true. Thus, the constraints imposed by voluntary participation are most severe when agents can withdraw at the ex post stage, and least severe when they can withdraw only at the ex ante stage.

In summary, when agents' types are privately observed, the set of social choice functions that can be successfully implemented are those that satisfy not only the conditions identified in Sections 23.C and 23.D for incentive compatibility (in, respectively, either a dominant strategy or Bayesian sense, depending on the equilibrium concept we employ) but also any participation constraints that are relevant in the environment under study.

In the remainder of this section, we illustrate further the limitations on the set of implementable social choice functions that may be caused by participation constraints by studying the important *Myerson–Satterthwaite theorem* [due to Myerson and Satterthwaite (1983)].

The Myerson–Satterthwaite Theorem

Consider again the bilateral trade setting introduced in Example 23.B.4. Agent 1 is the seller of an indivisible object and has a valuation for the object that lies in the interval $\Theta_1 = [\underline{\theta}_1, \bar{\theta}_1] \subset \mathbb{R}$; agent 2 is the buyer and has a valuation that lies in $\Theta_2 = [\underline{\theta}_2, \bar{\theta}_2] \subset \mathbb{R}$. The two valuations are statistically independent, and θ_i has distribution function $\Phi_i(\cdot)$ with an associated density $\phi_i(\cdot)$ satisfying $\phi_i(\theta_i) > 0$ for all $\theta_i \in [\underline{\theta}_i, \bar{\theta}_i]$. We let $y_i(\theta)$ denote the probability that agent i receives the good given types $\theta = (\theta_1, \theta_2)$, and so agent i's expected utility given θ is $\theta_i y_i(\theta) + t_i(\theta)$ (we normalize $\bar{m}_i = 0$ for all i).

The expected externality mechanism studied in Section 23.D shows that in this setting we can Bayesian implement an ex post efficient social choice function (or what, in this environment, we might call a "trading rule"). A problem arises with the expected externality mechanism, however, when trade is voluntary. In this case, every type of buyer and seller must have nonnegative expected gains from trade if he is to participate. In particular, if a seller of type θ_1 is to participate in a mechanism that implements social choice function $f(\cdot)$, that is, if participation in the mechanism is to be *individually rational* for this type of seller, it must be that $U_1(\theta_1 | f) \geq \theta_1$, because this seller can achieve an expected utility of θ_1 by not participating in the mechanism and simply consuming the good. Likewise, a buyer of type θ_2 can always earn zero by refusing to participate, and so we must have $U_2(\theta_2 | f) \geq 0$. Unfortunately, these interim participation constraints are not satisfied in the expected externality mechanism (you are asked to verify this in Exercise 23.E.1).

The *Myerson–Satterthwaite Theorem* tells us the following disappointing piece of

40. For example, if the mechanism can lead an agent into bankruptcy, the provisions of bankruptcy law provide an effective lower bound on ex post utilities.

news: Whenever gains from trade are possible, but are not certain,[41] there is *no* ex post efficient social choice function that is both Bayesian incentive compatible and satisfies these interim participation constraints. Thus, under the conditions of the theorem, the presence of both private information and voluntary participation implies that it is impossible to achieve ex post efficiency. (For an illustration of the result for specific functional forms, see Exercise 23.E.7.)

Proposition 23.E.1: (*The Myerson–Satterthwaite Theorem*) Consider a bilateral trade setting in which the buyer and seller are risk neutral, the valuations θ_1 and θ_2 are independently drawn from the intervals $[\underline{\theta}_1, \bar{\theta}_1] \subset \mathbb{R}$ and $[\underline{\theta}_2, \bar{\theta}_2] \subset \mathbb{R}$ with strictly positive densities, and $(\underline{\theta}_1, \bar{\theta}_1) \cap (\underline{\theta}_2, \bar{\theta}_2) \neq \varnothing$. Then there is no Bayesian incentive compatible social choice function that is ex post efficient and gives every buyer type and every seller type nonnegative expected gains from participation.

Proof: The argument consists of two steps:

Step 1: In any Bayesian incentive compatible and interim individually rational social choice function $f(\cdot) = [y_1(\cdot), y_2(\cdot), t_1(\cdot), t_2(\cdot)]$ in which $y_1(\theta_1, \theta_2) + y_2(\theta_1, \theta_2) = 1$ and $t_1(\theta_1, \theta_2) + t_2(\theta_1, \theta_2) = 0$, we must have

$$\int_{\underline{\theta}_1}^{\bar{\theta}_1} \int_{\underline{\theta}_2}^{\bar{\theta}_2} y_2(\theta_1, \theta_2)\left[\left(\theta_2 - \frac{1 - \Phi_2(\theta_2)}{\phi_2(\theta_2)}\right) - \left(\theta_1 + \frac{\Phi_1(\theta_1)}{\phi_1(\theta_1)}\right)\right] \phi_1(\theta_1)\phi_2(\theta_2)\, d\theta_2\, d\theta_1 \geq 0. \tag{23.E.4}$$

To see this, note first that the same argument that leads to (23.D.15) can be applied here to give [throughout the proof we suppress the argument f in $U_i(\theta_i | f)$ and simply write $U_i(\theta_i)$]:

$$E[-\bar{t}_2(\theta_2)] = \left[\int_{\underline{\theta}_1}^{\bar{\theta}_1} \int_{\underline{\theta}_2}^{\bar{\theta}_2} y_2(\theta_1, \theta_2)\left(\theta_2 - \frac{1 - \Phi_2(\theta_2)}{\phi_2(\theta_2)}\right)\phi_1(\theta_1)\phi_2(\theta_2)\, d\theta_2\, d\theta_1\right] - U_2(\underline{\theta}_2). \tag{23.E.5}$$

Also, because (23.D.12) implies that

$$U_1(\underline{\theta}_1) = U_1(\bar{\theta}_1) - \int_{\underline{\theta}_1}^{\bar{\theta}_1} \int_{\underline{\theta}_2}^{\bar{\theta}_2} y_1(\theta_1, \theta_2)\phi_2(\theta_2)\, d\theta_2\, d\theta_1,$$

condition (23.D.15) also implies that

$$E[-\bar{t}_1(\theta_1)] = \left[\int_{\underline{\theta}_1}^{\bar{\theta}_1} \int_{\underline{\theta}_2}^{\bar{\theta}_2} y_1(\theta_1, \theta_2)\left(\theta_1 + \frac{\Phi_1(\theta_1)}{\phi_1(\theta_1)}\right)\phi_1(\theta_1)\phi_2(\theta_2)\, d\theta_2\, d\theta_1\right] - U_1(\bar{\theta}_1). \tag{23.E.6}$$

Then, since $y_1(\theta_1, \theta_2) = 1 - y_2(\theta_1, \theta_2)$ we have

$$E[-\bar{t}_1(\theta_1)] = \left[\int_{\underline{\theta}_1}^{\bar{\theta}_1} \int_{\underline{\theta}_2}^{\bar{\theta}_2} \left(\theta_1 + \frac{\Phi_1(\theta_1)}{\phi_1(\theta_1)}\right)\phi_1(\theta_1)\phi_2(\theta_2)\, d\theta_2\, d\theta_1\right]$$

$$-\left[\int_{\underline{\theta}_1}^{\bar{\theta}_1} \int_{\underline{\theta}_2}^{\bar{\theta}_2} y_2(\theta_1, \theta_2)\left(\theta_1 + \frac{\Phi_1(\theta_1)}{\phi_1(\theta_1)}\right)\phi_1(\theta_1)\phi_2(\theta_2)\, d\theta_2\, d\theta_1\right] - U_1(\bar{\theta}_1).$$

41. That is, whenever $(\underline{\theta}_1, \bar{\theta}_1) \cap (\underline{\theta}_2, \bar{\theta}_2) \neq \varnothing$ (or equivalently, $\bar{\theta}_2 > \underline{\theta}_1$ and $\bar{\theta}_1 > \underline{\theta}_2$), so that for some realizations of $\theta = (\theta_1, \theta_2)$ there are gains from trade but for others there are not.

But

$$\left[\int_{\underline{\theta}_1}^{\bar{\theta}_1}\int_{\underline{\theta}_2}^{\bar{\theta}_2}\left(\theta_1+\frac{\Phi_1(\theta_1)}{\phi_1(\theta_1)}\right)\phi_1(\theta_1)\phi_2(\theta_2)\,d\theta_2\,d\theta_1\right]=\left[\int_{\underline{\theta}_1}^{\bar{\theta}_1}[\theta_1\phi_1(\theta_1)+\Phi_1(\theta_1)]\,d\theta_1\right]$$

$$=[\theta_1\Phi_1(\theta_1)]_{\underline{\theta}_1}^{\bar{\theta}_1}$$

$$=\bar{\theta}_1.$$

Thus,

$$E[-\bar{t}_1(\theta_1)]=\bar{\theta}_1-\left[\int_{\underline{\theta}_1}^{\bar{\theta}_1}\int_{\underline{\theta}_2}^{\bar{\theta}_2}y_2(\theta_1,\theta_2)\left(\theta_1+\frac{\Phi_1(\theta_1)}{\phi_1(\theta_1)}\right)\phi_1(\theta_1)\phi_2(\theta_2)\,d\theta_2\,d\theta_1\right]-U_1(\bar{\theta}_1). \tag{23.E.7}$$

Now, the fact that $t_1(\theta_1,\theta_2)+t_2(\theta_1,\theta_2)=0$ implies that $E[-t_1(\theta_1,\theta_2)]+E[-t_2(\theta_1,\theta_2)]=0$. So, adding (23.E.5) and (23.E.7) we see that

$$[U_1(\bar{\theta}_1)-\bar{\theta}_1]+U_2(\underline{\theta}_2)=$$

$$\int_{\underline{\theta}_1}^{\bar{\theta}_1}\int_{\underline{\theta}_2}^{\bar{\theta}_2}y_2(\theta_1,\theta_2)\left[\left(\theta_2-\frac{1-\Phi_2(\theta_2)}{\phi_2(\theta_2)}\right)-\left(\theta_1+\frac{\Phi_1(\theta_1)}{\phi_1(\theta_1)}\right)\right]\phi_1(\theta_1)\phi_2(\theta_2)\,d\theta_2\,d\theta_1.$$

But individual rationality implies that $U_1(\bar{\theta}_1)\geq\bar{\theta}_1$ and $U_2(\underline{\theta}_2)\geq 0$, which establishes (23.E.4).

Step 2: Condition (23.E.4) cannot be satisfied if $y_2(\theta_1,\theta_2)=1$ *whenever* $\theta_2>\theta_1$ *and* $y_2(\theta_1,\theta_2)=0$ *whenever* $\theta_2<\theta_1$.

Suppose it were. Then the left-hand side of (23.E.4) could be written as

$$\int_{\underline{\theta}_2}^{\bar{\theta}_2}\int_{\underline{\theta}_1}^{\mathrm{Min}\{\theta_2,\bar{\theta}_1\}}\left[\left(\theta_2-\frac{1-\Phi_2(\theta_2)}{\phi_2(\theta_2)}-\theta_1\right)\phi_1(\theta_1)-\Phi_1(\theta_1)\right]\phi_2(\theta_2)\,d\theta_1\,d\theta_2$$

$$=\int_{\underline{\theta}_2}^{\bar{\theta}_2}\left[\left(\theta_2-\frac{1-\Phi_2(\theta_2)}{\phi_2(\theta_2)}-\theta_1\right)\Phi_1(\theta_1)\right]_{\underline{\theta}_1}^{\mathrm{Min}\{\theta_2,\bar{\theta}_1\}}\phi_2(\theta_2)\,d\theta_2$$

$$=\int_{\underline{\theta}_2}^{\bar{\theta}_2}\left[\left(\theta_2-\frac{1-\Phi_2(\theta_2)}{\phi_2(\theta_2)}-\mathrm{Min}\{\theta_2,\bar{\theta}_1\}\right)\Phi_1(\mathrm{Min}\{\theta_2,\bar{\theta}_1\})\right]\phi_2(\theta_2)\,d\theta_2$$

$$=-\int_{\underline{\theta}_2}^{\bar{\theta}_1}[1-\Phi_2(\theta_2)]\Phi_1(\theta_2)\,d\theta_2+\int_{\bar{\theta}_1}^{\bar{\theta}_2}[(\theta_2-\bar{\theta}_1)\phi_2(\theta_2)+(\Phi_2(\theta_2)-1)]\,d\theta_2$$

$$=-\int_{\underline{\theta}_2}^{\bar{\theta}_1}[1-\Phi_2(\theta_2)]\Phi_1(\theta_2)\,d\theta_2+[(\theta_2-\bar{\theta}_1)(\Phi_2(\theta_2)-1)]_{\bar{\theta}_1}^{\bar{\theta}_2}$$

$$=-\int_{\underline{\theta}_2}^{\bar{\theta}_1}[1-\Phi_2(\theta_2)]\Phi_1(\theta_2)\,d\theta_2$$

$$<0,$$

where the inequality follows because $\bar{\theta}_1>\underline{\theta}_2$ and $\underline{\theta}_1<\bar{\theta}_2$. This contradicts (23.E.4) and completes the argument. ■

Recalling the revelation principle for Bayesian Nash equilibrium (Proposition 23.D.1), the implication of the Myerson–Satterthwaite theorem can be put as follows: Consider *any* voluntary trading institution that regulates trade between the buyer and the seller. This includes, for example, any bargaining process in which the parties can make offers and counteroffers to each other, as well as any arbitration mechanism in which the parties tell a third party their types and this third party then decides whether trade will occur and at what price.[42] By the revelation principle, we know that the social choice function that is indirectly implemented in a Bayesian Nash equilibrium[43] of such a mechanism must be Bayesian incentive compatible. Moreover, since participation is voluntary, this social choice function $f(\cdot)$ must satisfy the interim individual rationality constraints that $U_1(\theta_1 | f) \geq \theta_1$ for all θ_1 and $U_2(\theta_2 | f) \geq 0$ for all θ_2. Thus, the Myerson–Satterthwaite theorem tells us that, under its assumptions, no voluntary trading institution can have a Bayesian Nash equilibrium that leads to an ex post efficient outcome for all realizations of the buyer's and seller's valuations.

23.F Optimal Bayesian Mechanisms

In Sections 23.B to 23.E we have been concerned with the identification of implementable social choice functions in environments characterized by incomplete information about agents' preferences. In this section, we shift our focus to the welfare evaluaton of implementable social choice functions. We begin by developing several welfare criteria that extend the notion of Pareto efficiency that we have used throughout the book in the context of economies with complete information to these incomplete information settings. With these welfare notions in hand, we then discuss several examples that illustrate the characterization of optimal social choice functions (and, by implication, the optimal direct revelation mechanisms that implement them). We restrict our focus throughout this section to implementation in Bayesian Nash equilibria, discussed in detail in Section 23.D. Unless otherwise noted, we also adopt the assumptions and notation of Section 23.D. Good sources for further reading on the subject of this section are Holmstrom and Myerson (1983), Myerson (1991), and Fudenberg and Tirole (1991).

For economies in which agents' preferences are known with certainty, the concept of Pareto efficiency (or Pareto optimality) provides a minimal test that any welfare optimal outcome $x \in X$ should pass: There should be no other feasible outcome $\hat{x} \in X$ with the property that some agents are strictly better off with outcome $\hat{x}$ than with outcome x, and no agent is worse off.

The extension of this welfare test to social choice functions in settings of incomplete information should read something like the following:

> The social choice function $f(\cdot)$ is efficient if it is feasible and if there is no other feasible social choice function that makes some agents strictly better off, and no agents worse off.

To operationalize this idea, however, we need to be more specific about two things: First, what exactly do we mean by a social choice function being "feasible"? Second,

42. Strictly speaking, for a direct application of Proposition 23.E.1, the date of delivery and consumption of the good must be fixed (so the bargaining processes studied in Appendix A of Chapter 9 would not count). But through a suitable reinterpretation Proposition 23.E.1 can be applied to settings in which trade may take place over real time, where not only delivery of the good matters but also the *time of delivery* (see Exercise 23.E.4 for details).

43. And, hence, in any perfect Bayesian or sequential equilibrium (see Section 9.C).

precisely what do we mean when we say that no other feasible social choice function "makes some agents strictly better off, and no agent worse off"?

Let us consider the first of these issues. The identification of the set of feasible social choice functions when agents' preferences are private information has been discussed extensively in Sections 23.D and 23.E. Suppose that we define the set

$$F_{BIC} = \{f\colon \Theta \to X\colon f(\cdot) \text{ is Bayesian incentive compatible}\}. \tag{23.F.1}$$

The elements of set F_{BIC} in any particular application are the social choice functions that satisfy condition (23.D.1), the condition that assures that there is a Bayesian Nash equilibrium of the direct revelation mechanism $\Gamma = (\Theta_1, \ldots, \Theta_I, f(\cdot))$ in which truth telling is each agent's equilibrium strategy.

Likewise, following the discussion in Section 23.E, we can also define the set

$$F_{IR} = \{f\colon \Theta \to X\colon f(\cdot) \text{ is individually rational}\}. \tag{23.F.2}$$

The set F_{IR} contains those social choice functions that satisfy whichever of the three types of individual rationality (or participation) constraints (23.E.1)–(23.E.3) are relevant in the application being studied. If no individual rationality constraints are relevant (i.e., if agents' participation is not voluntary), then we simply have $F_{IR} = \{f\colon \Theta \to X\}$, the set of all possible social choice functions.

The content of our discussion in Sections 23.D and 23.E is therefore that the set of feasible social choice functions in environments in which agents' types are private information is precisely $F^* = F_{BIC} \cap F_{IR}$. Following Myerson (1991), we call this the *incentive feasible set* to emphasize that it is the set of feasible social choice functions when, because of incomplete information, incentive compatibility conditions must be satisfied.

Now consider the second issue: What do we mean when we say that no other feasible social choice function would "make some agents strictly better off, and no agents worse off"? The critical issue here has to do with the *timing* of our welfare analysis. In particular, is the welfare analysis occurring *before* the agents (privately) learn their types, or *after*? The former amounts to a welfare analysis conducted at what we called in Section 23.E the *ex ante stage* (the point in time at which agents have not yet learned their types); the latter corresponds to what we called in Section 23.E the *interim stage* (the point in time after each agent has learned his type, but before the agents' types are publicly revealed). To formally define the different welfare criteria that arise in these two cases, let us once again denote by $U_i(\theta_i|f)$ agent i's expected utility from social choice function $f(\cdot)$ conditional on being of type θ_i. Also let $U_i(f) = E_{\theta_i}[U_i(\theta_i|f)]$ denote agent i's ex ante expected utility from social choice function $f(\cdot)$. We can now state Definitions 23.F.1 and 23.F.2.

Definition 23.F.1: Given any set of feasible social choice functions F, the social choice function $f(\cdot) \in F$ is *ex ante efficient in F* if there is no $\hat{f}(\cdot) \in F$ having the property that $U_i(\hat{f}) \geq U_i(f)$ for all $i = 1, \ldots, I$, and $U_i(\hat{f}) > U_i(f)$ for some i.

Definition 23.F.2: Given any set of feasible social choice functions F, the social choice function $f(\cdot) \in F$ is *interim efficient in F* if there is no $\hat{f}(\cdot) \in F$ having the property that $U_i(\theta_i|\hat{f}) \geq U_i(\theta_i|f)$ for all $\theta_i \in \Theta_i$ and all $i = 1, \ldots, I$, and $U_i(\theta_i|\hat{f}) > U_i(\theta_i|f)$ for some i and $\theta_i \in \Theta_i$.

The motivation for the ex ante efficiency test is straightforward: If agents have not yet learned their types, then when comparing two feasible social choice functions we should evaluate each agent's well-being using his expected utility over all of his possible types. However, when our welfare analysis occurs after agents have (privately) learned their types, things are a bit trickier. Although the agents each know their types, we—as outsiders—do not know them. Thus, the appropriate notion for us to adopt in saying that one social choice function $\hat{f}(\cdot)$ welfare dominates another social choice function $f(\cdot)$ is that $\hat{f}(\cdot)$ makes every possible type of every agent at least as well off as does $f(\cdot)$, and makes some type of some agent strictly better off. This leads to the concept of interim efficiency given in Definition 23.F.2.

Proposition 23.F.1 compares these two notions of efficiency.

Proposition 23.F.1: Given any set of feasible social choice functions F, if the social choice function $f(\cdot) \in F$ is ex ante efficient in F, then it is also interim efficient in F.

Proof: Suppose that $f(\cdot)$ is ex ante efficient in F but is not interim efficient in F. Then there exists an $\hat{f}(\cdot) \in F$ such that $U_i(\theta_i \mid \hat{f}) \geq U_i(\theta_i \mid f)$ for all $\theta_i \in \Theta_i$ and all $i = 1, \ldots, I$, and $U_i(\theta_i \mid \hat{f}) > U_i(\theta_i \mid f)$ for some i and $\theta_i \in \Theta_i$. But since, for all i, $U_i(f) = E_{\theta_i}[U_i(\theta_i \mid f)]$ and $U_i(\hat{f}) = E_{\theta_i}[U_i(\theta_i \mid \hat{f})]$, it follows that $U_i(\hat{f}) \geq U_i(f)$ for all $i = 1, \ldots, I$, and $U_i(\hat{f}) > U_i(f)$ for some i, contradicting the hypothesis that $f(\cdot)$ is ex ante efficient in F. ■

The ex ante efficiency concept is more demanding than is interim efficiency (and so fewer social choice functions $f(\cdot)$ pass the ex ante efficiency test) because a social choice function $\hat{f}(\cdot)$ can raise every agent's ex ante expected utility relative to the social choice function $f(\cdot)$ even though $\hat{f}(\cdot)$ may lead some type of some agent i to have a lower expected utility than he does with $f(\cdot)$.

Putting together the elements developed above, we conclude that when agents' types are already determined at the time we are conducting our welfare analysis, the proper notion of efficiency of a social choice function in an environment with incomplete information is interim efficiency in F^*, the set of Bayesian incentive compatible and individually rational social choice functions.[44] On the other hand, if our analysis is conducted prior to agents learning their types, then the proper notion of efficiency is ex ante efficiency in F^*.[45] These two notions are often called simply *ex ante incentive efficiency* and *interim incentive efficiency* [the terminology is due to Holmstrom and Myerson (1983)], where the modifier "incentive" is meant to convey the point that the set F^* is being used.[46]

These two welfare notions differ from the ex post efficiency criterion introduced in Definition 23.B.2. To see their relationship to it more clearly, Definition 23.F.3

44. These cases often correspond to situations in which our assumption that the agents' types are drawn from a known prior distribution is being used merely as a device to model agents' beliefs about each others' types, as described in Section 8.E, rather than as a description of any actual prior time at which the agents could interact or our welfare analysis might have been done.

45. This case often arises in contracting problems when, at the time of contracting, the agents anticipate that they will later come to acquire private information about their types. Then the natural welfare standard to use in comparing different contracts (i.e., different mechanisms) is the ex ante criterion. The principal–agent model studied in Section 14.C and Example 23.F.1 below is an example along these lines.

46. However, since the relevant individual rationality constraints vary from one application to another, it is usually clearer to describe precisely the set F within which efficiency is being evaluated.

develops the ex post efficiency notion in a manner that parallels Definitions 23.F.1 and 23.F.2.

Definition 23.F.3: Given any set of feasible social choice functions F, the social choice function $f(\cdot) \in F$ is *ex post efficient in F* if there is no $\hat{f}(\cdot) \in F$ having the property that $u_i(\hat{f}(\theta), \theta_i) \geq u_i(f(\theta), \theta_i)$ for all $i = 1, \ldots, I$ and all $\theta \in \Theta$, and $u_i(\hat{f}(\theta), \theta_i) > u_i(f(\theta), \theta_i)$ for some i and $\theta \in \Theta$.

The ex post efficiency test in Definition 23.F.3 conducts its welfare evaluation at the ex post stage at which all agents' information has been publicly revealed. Using this definition, we see that a social choice function $f(\cdot)$ is ex post efficient in the sense of Definition 23.B.2 if and only if it is ex post efficient in the sense of Definition 23.F.3 when we take $F = \{f: \Theta \to X\}$.

Note that the criterion of ex post efficiency in $\{f: \Theta \to X\}$, or more generally, of ex post efficiency in F_{IR} when individual rationality constraints are present, ignores issues of incentive compatibility. As a result, it is appropriate as a welfare criterion only if agents' types are in fact publicly observable. Because $F^* \subset F_{IR}$, allocations that are ex ante or interim incentive efficient need not be ex post efficient in this sense. Indeed, the Myerson–Satterthwaite theorem (Proposition 23.E.1) provides an illustration of this phenomenon for the bilateral trade setting: under its assumptions, no element of F^* is ex post efficient. Examples 23.F.1 to 23.F.3 provide further illustrations. (For one way in which the notion of ex post efficiency is nevertheless still of interest in settings with privately observed types, see Exercise 23.F.1.)

Note also that even in settings in which agents' types are public information, the use of ex post efficiency in F_{IR} as our welfare criterion is appropriate only when agents' types are already determined. When our welfare analysis instead occurs prior to agents learning their types, the appropriate notion is instead the stronger criterion that $f(\cdot)$ be ex ante efficient in F_{IR}. These two notions are sometimes called *ex post classical efficiency* and *ex ante classical efficiency* [again, the terminology is due to Holmstrom and Myerson (1983)] to indicate that no incentive constraints are involved in defining the feasible set of social choice functions.

In the remainder of this section we study three examples in which we characterize welfare optimal social choice functions. In Examples 23.F.1 and 23.F.2, it is supposed that one agent who receives no private information chooses a mechanism to maximize his expected utility subject to both incentive compatibility constraints and interim individual rationality constraints for the other agents. These two examples therefore amount to a characterization of one particular interim incentive efficient mechanism. In Example 23.F.3, we provide a full characterization of the sets of interim and ex ante incentive efficient social choice functions for a simple setting of bilateral trade with adverse selection.

Example 23.F.1: *A Principal–Agent Problem with Hidden Information.* In Section 14.C we studied principal–agent problems with hidden information for the case in which the agent has two possible types. Here we consider the case where the agent may have a continuum of types. Recall from Section 14.C that in the principal–agent problem with hidden information, the principal faces the problem of designing an optimal (i.e., payoff maximizing) contract for an agent who will come to possess private information. In doing so, the principal faces both incentive constraints and a reservation utility constraint for the agent. Recall also from Section 14.C that, in the limiting case in which the agent is infinitely risk averse, the agent must be guaranteed his reservation utility for each possible type he may come to have, and so this contracting problem is identical to the contracting problem that would arise if the agent already knew his type at the time of contracting. Here we shall set things up directly in these terms, assuming that the agent already possesses this information when contracting occurs. With this formulation, the principal's optimal contract can be viewed as implementing one particular interim incentive efficient social choice function. (When the agent actually does not know his type at the time of contracting and is infinitely risk averse, then this social choice function is also ex ante incentive efficient.)

To introduce our notation, we suppose that the agent (individual 1) may take some observable action $e \in \mathbb{R}_+$ (his "effort" or "task" level) and receives a monetary payment from the principal of t_1. The agent's type is drawn from the interval $[\underline{\theta}, \bar{\theta}]$, where $\underline{\theta} < \bar{\theta} < 0$, according to the distribution function $\Phi(\cdot)$ which has an associated density function $\phi(\cdot)$ that is strictly positive on $[\underline{\theta}, \bar{\theta}]$. We assume that this distribution satisfies the property that $[\theta - ((1 - \Phi(\theta))/\phi(\theta))]$ is nondecreasing in θ.[47]

The agent's Bernoulli utility function when his type is θ is $u_1(e, t_1, \theta) = t_1 + \theta g(e)$, where $g(\cdot)$ is a differentiable function with $g(0) = 0$, $g(e) > 0$ for $e > 0$, $g'(0) = 0$, $g'(e) > 0$ for $e > 0$, and $g''(\cdot) > 0$; that is, $\theta g(e)$ represents the agent's disutility of effort (recall that $\theta < 0$), with higher effort levels leading to an increasing level of disutility to the agent. Note that a larger (i.e., less negative) level of θ lowers, at any level of e, both the agent's total level of disutility and his marginal disutility from any increase in e. As noted above, we suppose that the agent must be guaranteed an expected utility level of at least $\bar{u}$ for each possible type he may have.

The principal (individual 0) has no private information. His Bernoulli utility function is $u_0(e, t_0) = v(e) + t_0$, where t_0 is his net transfer and $v(\cdot)$ is a differentiable function satisfying $v'(\cdot) > 0$ and $v''(\cdot) < 0$.

A contract between the principal and the agent can be viewed as specifying a mechanism in the sense we have used throughout this chapter. By the revelation principle for Bayesian Nash equilibrium (Proposition 23.D.1), the equilibrium outcome induced by such a contract, formally a social choice function that maps each possible agent type into effort and transfer levels, can always be duplicated using a direct revelation mechanism that induces truth telling. Thus, the principal can confine his search for an optimal contract to the set of Bayesian incentive compatible social choice functions $f(\cdot) = (e(\cdot), t_0(\cdot), t_1(\cdot))$ that give the agent an expected utility of at least $\bar{u}$ for every possible value of θ. In what follows, we shall (without loss of generality) restrict attention in our search for the principal's optimal contract to contracts that have $t_0(\theta) = -t_1(\theta)$ for all θ (i.e., that involve no waste of numeraire).

The principal's problem can therefore be stated as

$$\begin{aligned} \max_{f(\cdot)=(e(\cdot),t_1(\cdot))} \quad & E[v(e(\theta)) - t_1(\theta)] \\ \text{s.t. } & f(\cdot) \text{ is Bayesian incentive compatible and} \\ & \text{individually rational.} \end{aligned}$$

47. For a discussion of how the analysis changes when this assumption is not satisfied, see Fudenberg and Tirole (1991).

The present model falls into the class of models with linear utility studied in Section 23.D [specifically, in the notation of Proposition 23.D.2, $k = e$, $v_1(k) = g(e)$, and $\bar{v}_1(\theta) = g(e(\theta))$ here]. Letting $U_1(\theta) = t_1(\theta) + \theta g(e(\theta))$ denote the agent's utility if his type is θ and he tells the truth, Proposition 23.D.2 can be used to restate the principal's problem in terms of choosing the functions $e(\cdot)$ and $U_1(\cdot)$ to solve

$$\begin{aligned} \underset{e(\cdot),\, U_1(\cdot)}{\text{Max}} \quad & E[v(e(\theta)) + \theta g(e(\theta)) - U_1(\theta)] \\ \text{s.t.} \quad & \text{(i) } e(\cdot) \text{ is nondecreasing} \\ & \text{(ii) } U_1(\theta) = U_1(\underline{\theta}) + \int_{\underline{\theta}}^{\theta} g(e(s))\, ds \text{ for all } \theta \\ & \text{(iii) } U_1(\theta) \geq \bar{u} \text{ for all } \theta. \end{aligned} \tag{23.F.3}$$

Constraints (i) and (ii) are the necessary and sufficient conditions for the principal's contract to be Bayesian incentive compatible, adapted from Proposition 23.D.2 [constraint (i) follows because $g(\cdot)$ is increasing in e], while constraint (iii) is the agent's individual rationality constraint.

Note first that if constraint (ii) is satisfied, then constraint (iii) will be satisfied if and only if $U_1(\underline{\theta}) \geq \bar{u}$. As a result, we can replace constraint (iii) with

$$\text{(iii}'\text{)} \quad U_1(\underline{\theta}) \geq \bar{u}.$$

Next, substituting for $U_1(\theta)$ in the objective function from constraint (ii), and then integrating by parts in a fashion similar to the steps leading to (23.D.14), problem (23.F.3) can be restated as

$$\begin{aligned} \underset{e(\cdot),\, U_1(\underline{\theta})}{\text{Max}} \quad & \left[\int_{\underline{\theta}}^{\bar{\theta}} \left\{ v(e(\theta)) + g(e(\theta)) \left(\theta - \frac{1 - \Phi(\theta)}{\phi(\theta)} \right) \right\} \phi(\theta)\, d\theta \right] - U_1(\underline{\theta}) \\ \text{s.t.} \quad & \text{(i) } e(\cdot) \text{ is nondecreasing} \\ & \text{(iii}'\text{) } U_1(\underline{\theta}) \geq \bar{u}. \end{aligned} \tag{23.F.4}$$

It is now immediate from (23.F.4) that in any solution we must in fact have $U_1(\underline{\theta}) = \bar{u}$. Thus, we can write the principal's problem as one of choosing $e(\cdot)$ to solve

$$\begin{aligned} \underset{e(\cdot)}{\text{Max}} \quad & \left[\int_{\underline{\theta}}^{\bar{\theta}} \left\{ v(e(\theta)) + g(e(\theta)) \left(\theta - \frac{1 - \Phi(\theta)}{\phi(\theta)} \right) \right\} \phi(\theta)\, d\theta \right] - \bar{u} \\ \text{s.t.} \quad & \text{(i) } e(\cdot) \text{ is nondecreasing.} \end{aligned} \tag{23.F.5}$$

Suppose for the moment that we can ignore constraint (i). Then the optimal function $e(\cdot)$ must satisfy the first-order condition[48]

$$v'(e(\theta)) + g'(e(\theta)) \left(\theta - \frac{1 - \Phi(\theta)}{\phi(\theta)} \right) = 0 \qquad \text{for all } \theta. \tag{23.F.6}$$

But note that, under our assumption that $[\theta - ((1 - \Phi(\theta))/\phi(\theta))]$ is nondecreasing in θ, the implicit function theorem applied to (23.F.6) tells us that any solution $e(\cdot)$ to this relaxed problem must in fact be nondecreasing. Thus, (23.F.6) characterizes the solution to the principal's actual problem (see Section M.K of the Mathematical Appendix). The optimal $U_1(\cdot)$ [and, hence, $t_1(\cdot)$] is then calculated from constraint (ii) of (23.F.3) using this optimal $e(\cdot)$ and the fact that $U_1(\underline{\theta}) = \bar{u}$.

48. It can be shown that under our assumptions, the optimal contract is interior, that is, has $e(\theta) > 0$ for (almost) all θ.

It is interesting to compare this solution with the optimal contract for the case in which the agent's type is observable. This contract solves

$$\begin{aligned} \underset{e(\cdot),t_1(\cdot)}{\text{Max}} \quad & E[v(e(\theta)) - t_1(\theta)] \\ & \text{s.t. } t_1(\theta) + \theta g(e(\theta)) \geq \bar{u} \text{ for all } \theta. \end{aligned}$$

Hence, the optimal task level in this complete information contract is the level $e^*(\theta)$ that satisfies, for all θ,

$$v'(e^*(\theta)) + g'(e^*(\theta))\theta = 0.$$

Note that $e^*(\theta)$ is the level that arises in any ex post (classically) efficient social choice function. In contrast, the principal's optimal $e(\cdot)$ when θ is private information is such that

$$v'(e(\theta)) + g'(e(\theta))\theta \begin{cases} > 0 & \text{at all } \theta < \bar{\theta}, \\ = 0 & \text{at } \theta = \bar{\theta}. \end{cases}$$

We see then that $e(\theta) < e^*(\theta)$ for all $\theta < \bar{\theta}$, and $e(\bar{\theta}) = e^*(\bar{\theta})$. This is a version of the same result that we saw for the two-type case in Section 14.C. In the optimal contract, the type of agent with the lowest disutility from effort (here type $\bar{\theta}$; in Section 14.C, type θ_H) takes an ex post efficient action, while all other types have their effort levels distorted downward. The reason is also the same: doing so helps reduce the amount the agent's utility exceeds his reservation utility for types $\theta > \underline{\theta}$ (his so-called *information rents*). To see this point heuristically, suppose that starting with some function $e(\cdot)$ we lower $e(\hat{\theta})$ by an amount $de < 0$ for some type $\hat{\theta} \in (\underline{\theta}, \bar{\theta})$ and lower this type's transfer to keep his utility unchanged.[49] The decrease in the transfer paid to type $\hat{\theta}$ is $\hat{\theta} g'(e(\hat{\theta}))\, de$, while the direct effect on the principal is $v'(e(\hat{\theta}))\, de$. At the same time, according to constraint (ii), this change in $e(\hat{\theta})$ lowers the utility level, and hence the transfer, that must be given to all types $\theta > \hat{\theta}$ by exactly $g'(e(\hat{\theta}))\, de$. The expected value of this reduction in the transfers paid to these types is $-(1 - \Phi(\hat{\theta}))g'(e(\hat{\theta}))\, de$. If the original $e(\cdot)$ is an optimum, the sum of the first two changes in the principal's profits (those for type $\hat{\theta}$) weighted by the density of type $\hat{\theta}$, $[v'(e(\hat{\theta})) + \hat{\theta} g'(e(\hat{\theta}))]\phi(\hat{\theta})\, de$, plus the reduction in payments to types $\theta > \hat{\theta}$, $(1 - \Phi(\hat{\theta}))g'(e(\hat{\theta}))\, de$, must equal zero. This gives exactly (23.F.6). ■

Example 23.F.2: *Optimal Auctions.* We consider again the auction setting introduced in Example 23.B.4. Here we determine the optimal auction for the seller of an indivisible object (agent 0) when there are I buyers, indexed by $i = 1, \dots, I$. Each buyer has a Bernoulli utility function $\theta_i y_i(\theta) + t_i(\theta)$, where $y_i(\theta)$ is the probability that agent i gets the good when the agents' types are $\theta = (\theta_1, \dots, \theta_I)$. In addition, each buyer i's type is independently drawn according to the distribution function $\Phi_i(\cdot)$ on $[\underline{\theta}_i, \bar{\theta}_i] \subset \mathbb{R}$ with $\underline{\theta}_i \neq \bar{\theta}_i$ and associated density $\phi_i(\cdot)$ that is strictly positive on $[\underline{\theta}_i, \bar{\theta}_i]$. We assume also that, for $i = 1, \dots, I$,

$$\theta_i - \frac{1 - \Phi_i(\theta_i)}{\phi_i(\theta_i)}$$

is nondecreasing in θ_i.[50]

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49. We say "heuristically" because to do this rigorously we need to perform this reduction in e over an interval of types and then take limits.

50. For a discussion of the case in which this assumption is not met, see Myerson (1981).

A social choice function in this environment is a function $f(\cdot) = (y_0(\cdot), \ldots, y_I(\cdot), t_0(\cdot), \ldots, t_I(\cdot))$ having the properties that, for all $\theta \in \Theta$, $y_i(\theta) \in [0, 1]$ for all i, $\sum_{i \neq 0} y_i(\theta) = 1 - y_0(\theta)$, and $t_0(\theta) = -\sum_{i \neq 0} t_i(\theta)$.[51] The seller wishes to choose the Bayesian incentive compatible social choice function that maximizes his expected revenue $E_\theta[t_0(\theta)] = -E_\theta[\sum_{i \neq 0} t_i(\theta)]$ but faces the interim individual rationality constraints that $U_i(\theta_i) = \theta_i \bar{y}_i(\theta_i) + \bar{t}_i(\theta_i) \geq 0$ for all θ_i and $i \neq 0$ [as in Section 23.D, $\bar{y}_i(\theta_i)$ and $\bar{t}_i(\theta_i)$ are agent i's probability of getting the good and expected transfer conditional on announcing his type to be θ_i] because buyers are always free not to participate. The seller's optimal choice is therefore a particular element of the set of interim incentive efficient social choice functions.

The seller's problem can be written as one of choosing functions $y_1(\cdot), \ldots, y_I(\cdot)$ and $U_1(\cdot), \ldots, U_I(\cdot)$ to solve

$$\underset{\{y_i(\cdot), U_i(\cdot)\}_{i=1}^I}{\text{Max}} \sum_{i \neq 0} \int_{\underline{\theta}_i}^{\bar{\theta}_i} [\bar{y}_i(\theta_i)\theta_i - U_i(\theta_i)]\phi_i(\theta_i)\, d\theta_i \tag{23.F.7}$$

s.t. (i) $\bar{y}_i(\cdot)$ is nondecreasing for all $i \neq 0$.

(ii) For all θ: $y_i(\theta) \in [0, 1]$ for all $i \neq 0$, $\sum_{i \neq 0} y_i(\theta) \leq 1$.

(iii) $U_i(\theta_i) = U_i(\underline{\theta}_i) + \int_{\underline{\theta}_i}^{\theta_i} \bar{y}_i(s)\, ds$ for all $i \neq 0$ and θ_i.

(iv) $U_i(\theta_i) \geq 0$ for all $i \neq 0$ and θ_i.

We note first that if constraint (iii) is satisfied then constraint (iv) will be satisfied if and only if $U_i(\underline{\theta}_i) \geq 0$ for all $i \neq 0$. As a result, we can replace constraint (iv) with

(iv') $U_i(\underline{\theta}_i) \geq 0$ for all $i \neq 0$ and θ_i.

Next, substituting into the objective function for $U_i(\theta_i)$ using constraint (iii), and following the same steps that led to (23.D.16), the seller's problem can be written as one of choosing the $y_i(\cdot)$ functions and the values $U_1(\underline{\theta}_1), \ldots, U_I(\underline{\theta}_I)$ to maximize

$$\int_{\underline{\theta}_1}^{\bar{\theta}_1} \cdots \int_{\underline{\theta}_I}^{\bar{\theta}_I} \left[\sum_{i=1}^{I} y_i(\theta_1, \ldots, \theta_I)\left(\theta_i - \frac{1 - \Phi_i(\theta_i)}{\phi_i(\theta_i)}\right)\right]\left[\prod_{i=1}^{I} \phi_i(\theta_i)\right] d\theta_I \cdots d\theta_1 - \sum_{i=1}^{I} U_i(\underline{\theta}_i)$$

subject to constraints (i), (ii), and (iv'). It is evident that the solution must have $U_i(\underline{\theta}_i) = 0$ for all $i = 1, \ldots, I$. Hence, the seller's problem reduces to choosing functions $y_1(\cdot), \ldots, y_I(\cdot)$ to maximize

$$\int_{\underline{\theta}_1}^{\bar{\theta}_1} \cdots \int_{\underline{\theta}_I}^{\bar{\theta}_I} \left[\sum_{i=1}^{I} y_i(\theta_1, \ldots, \theta_I)\left(\theta_i - \frac{1 - \Phi_i(\theta_i)}{\phi_i(\theta_i)}\right)\right]\left[\prod_{i=1}^{I} \phi_i(\theta_i)\right] d\theta_I \cdots d\theta_1 \tag{23.F.8}$$

subject to constraints (i) and (ii).

Let us ignore constraint (i) for the moment. Define

$$J_i(\theta_i) = \theta_i - \frac{1 - \Phi_i(\theta_i)}{\phi_i(\theta_i)}.$$

Then inspection of (23.F.8) indicates that $y_1(\cdot), \ldots, y_I(\cdot)$ is a solution to this relaxed

51. Once again we restrict attention, without loss of generality, to social choice functions involving no waste of either the numeraire or the good (there is always an optimal social choice function for the seller with this form).

problem if and only if for all $i = 1, \dots, I$ we have

$$y_i(\theta) = 1 \quad \text{if } J_i(\theta_i) > \text{Max}\,\{0, \text{Max}_{h \neq i} J_h(\theta_h)\}$$

and (23.F.9)

$$y_i(\theta) = 0 \quad \text{if } J_i(\theta_i) < \text{Max}\,\{0, \text{Max}_{h \neq i} J_h(\theta_h)\}.$$

[Note that $J_i(\theta_i) = \text{Max}\,\{0, \text{Max}_{h \neq i} J_h(\theta_h)\}$ is a zero probability event.] But, given our assumption that $J_i(\cdot)$ is nondecreasing in θ_i, (23.F.9) implies that $y_i(\cdot)$ is nondecreasing in θ_i, which in turn implies that $\bar{y}_i(\cdot)$ is nondecreasing. Thus the solution to this relaxed problem actually satisfies constraint (i), and so is a solution to the seller's overall problem (see Section M.K of the Mathematical Appendix). The optimal transfer functions can then be set as $t_i(\theta) = U_i(\theta_i) - \theta_i \bar{y}_i(\theta_i)$, where $U_i(\theta_i)$ is calculated from constraint (iii).

A few things should be noted about (23.F.9). First, observe that when the various agents have differing distribution functions $\Phi_i(\cdot)$, the agent i who has the largest value of $J_i(\theta_i)$ is *not* necessarily the same as the agent who has the highest valuation for the object. Thus, the seller's optimal auction need not be ex post (classically) efficient.

Second, in the case of *symmetric* bidders in which $\underline{\theta}_i = \underline{\theta}$ and $J_i(\cdot) = J(\cdot)$ for $i = 1, \dots, I$, when $\underline{\theta} > 0$ is large enough so that $J(\underline{\theta}) > 0$, the optimal auction always gives the object to the bidder with the highest valuation and also leaves each bidder with an expected utility of zero when his valuation attains its lowest possible value. We can therefore conclude, using the revenue equivalence theorem (Proposition 23.D.3), that the first-price and second-price sealed-bid auctions are both optimal in this case.

Third, the optimal auction has a nice interpretation in terms of monopoly pricing. Consider, for example, the case in which $I = 1$ and $\underline{\theta}_1 = 0$. Then conditions (23.F.9) tell us that the optimal auction gives the object to the buyer (agent 1) if and only if $J_1(\theta_1) = [\theta_1 - ((1 - \Phi_1(\theta_1))/\phi_1(\theta_1))] > 0$. Suppose we think instead of the seller in this circumstance simply naming a price p and letting the buyer then decide whether to buy at this price. The seller's expected revenue from this scheme is $p(1 - \Phi_1(p))$, and so the first-order condition for his optimal posted price, say p^*, is $(1 - \Phi_1(p^*)) - p^*\phi_1(p^*) = 0$, or equivalently, $J_1(p^*) = 0$. Since $J_1(\cdot)$ is nondecreasing, we see that with this optimal posted price policy a buyer of type θ_1 gets the good with probability 1 if $J_1(\theta_1) > 0$, and with probability 0 if $J_1(\theta_1) < 0$, exactly as in the optimal auction derived above. Indeed, given the revenue equivalence theorem we can conclude that in this case this simple posted price scheme is an optimal mechanism for the seller. [For more on the monopoly interpretation of optimal auctions, see Exercise 23.F.5 and Bulow and Roberts (1989).] ■

Throughout the chapter, we have restricted attention to "private values" settings in which agents' utilities depend only on their own types. In a number of settings of economic interest, however, an agent i's utility depends not only his own type, θ_i, but also on the types of other agents, θ_{-i}. That is, agent i's Bernoulli utility function may take the form $u_i(x, \theta)$ rather than $u_i(x, \theta_i)$, where $\theta = (\theta_i, \theta_{-i})$. Fortunately, all of the concepts of implementation for Bayesian Nash equilibrium that we have studied in Sections 23.D to 23.F extend readily to this case. For example, we can say that the social choice function $f\colon \Theta \to X$ is Bayesian incentive compatible if for all i

and all $\theta_i \in \Theta_i$,

$$E_{\theta_{-i}}[u_i(f(\theta_i, \theta_{-i}), \theta) \mid \theta_i] \geq E_{\theta_{-i}}[u_i(f(\hat{\theta}_i, \theta_{-i}), \theta) \mid \theta_i] \qquad (23.F.10)$$

for all $\hat{\theta}_i \in \Theta_i$. Our third and last example, which studies a simple bilateral trade setting with adverse selection (see Section 13.B for more on adverse selection), falls within this class of models. Another difference from the analysis of Examples 23.F.1 and 23.F.2 is that here we shall characterize the *entire sets* of both ex ante and interim incentive efficient social choice functions.

Example 23.F.3: *Bilateral Trade with Adverse Selection* [from Myerson (1991)]. Consider a bilateral trade setting in which there is a seller (agent 1) and a potential buyer (agent 2) of one unit of an indivisible private good. The good may be of high quality or low quality, but only the seller observes which is the case. To model this, we let the seller have two possible types, so that $\Theta_1 = \{\theta_L, \theta_H\}$, and we assume that $\text{Prob}(\theta_H) = .2$. Both the buyer's and the seller's utilities from consumption of the good depend on the seller's type. In particular, letting y denote the probability that the buyer receives the good, and letting t denote the amount of any monetary transfer from the buyer to the seller, we suppose that[52]

$$\begin{aligned} &u_1(y, t \mid \theta_L) = t + 20(1 - y), \qquad &u_1(y, t \mid \theta_H) = t + 40(1 - y), \\ &u_2(y, t \mid \theta_L) = 30y - t, \qquad &u_2(y, t \mid \theta_H) = 50y - t. \end{aligned} \qquad (23.F.11)$$

A social choice function in this setting assigns a probability of trade and a transfer for each possible value of θ_1, and so can be represented by a vector (y_L, t_L, y_H, t_H). We suppose that trade is voluntary, and that as a result any feasible social choice function must satisfy interim individual rationality constraints for both the buyer and the seller. For the seller this means that, for each type he may have, his expected utility must be no less than his utility from refusing to participate and simply consuming the good. Hence, we must have

$$t_L + 20(1 - y_L) \geq 20 \qquad (\text{IR}_{1L}) \qquad (23.F.12)$$

$$t_H + 40(1 - y_H) \geq 40. \qquad (\text{IR}_{1H}) \qquad (23.F.13)$$

For the buyer, on the other hand, interim individual rationality simply requires that he receive nonnegative expected utility from participation (recall that he does not observe θ_1). Hence, we must have

$$.2(50y_H - t_H) + .8(30y_L - t_L) \geq 0. \qquad (\text{IR}_2) \qquad (23.F.14)$$

Note from (23.F.11) that if θ_1 were publicly observable then, for each value of θ_1, there would be gains from trade between the buyer and the seller. Because of this fact, any ex post (classically) efficient social choice function has $y_H = y_L = 1$ (see Exercise 23.F.8).

Because θ_1 is only privately observed, the set of feasible social choice functions is the incentive feasible set F^*, the set of Bayesian incentive compatible and (interim) individually rational social choice functions. In the present context, the social choice

52. We assume that there is no waste of either the good or the numeraire, so that the probability that either the buyer or the seller consumes the good is 1, and any transfer from the buyer goes to the seller.

function (y_L, t_L, y_H, t_H) is Bayesian incentive compatible if

$$t_H + 40(1 - y_H) \geq t_L + 40(1 - y_L) \qquad (\text{IC}_H) \qquad (23.\text{F}.15)$$

and

$$t_L + 20(1 - y_L) \geq t_H + 20(1 - y_H). \qquad (\text{IC}_L) \qquad (23.\text{F}.16)$$

Condition (23.F.15) requires that truth telling be an optimal strategy for the seller when his type is θ_H; (23.F.16) is the condition for truth telling to be optimal when his type is θ_L. Thus, (y_L, t_L, y_H, t_H) is a feasible social choice function if and only if it satisfies the incentive compatibility constraints (23.F.15)–(23.F.16) and the interim individual rationality constraints (23.F.12)–(23.F.14).

These constraints imply that *any* feasible social choice function possesses the following three properties (Exercise 23.F.9 asks you to establish these points):

(i) No feasible social choice function is ex post (classically) efficient.
(ii) In any feasible social choice function, $y_H \leq y_L$ and $t_H \leq t_L$.
(iii) In any feasible social choice function, the expected gains from trade for a low-quality seller are at least as large as the expected gains from trade for a high-quality seller, that is, $t_L - 20y_L \geq t_H - 40y_H$.

We now proceed to characterize the interim and ex ante incentive efficient social choice functions for this bilateral trade problem. To determine the interim incentive efficient social choice functions, we need to determine the (y_L, t_L, y_H, t_H) that solve, for each possible choice of $\bar{u}_{1H} \geq 0$ and $\bar{u}_2 \geq 0$, the following problem (we have simplified the incentive compatibility and individual rationality constraints by eliminating constants on both sides of the inequalities, and have removed a constant from the objective function as well[53]):

$$\begin{aligned} \underset{(y_L \in [0,1], t_L, y_H \in [0,1], t_H)}{\text{Max}} \quad & t_L - 20y_L & (23.\text{F}.17) \\ \text{s.t.} \quad & \text{(i)} \ t_H - 40y_H \geq t_L - 40y_L \\ & \text{(ii)} \ t_L - 20y_L \geq t_H - 20y_H \\ & \text{(iii)} \ t_H - 40y_H \geq \bar{u}_{1H} \\ & \text{(iv)} \ .2(50y_H - t_H) + .8(30y_L - t_L) \geq \bar{u}_2. \end{aligned}$$

Problem (23.F.17) characterizes interim incentive efficient social choice functions by maximizing the interim expected utility of the type θ_L seller subject to giving the type θ_H seller an interim expected utility of at least $\bar{u}_{1H} \geq 0$, giving the buyer an interim expected utility of $\bar{u}_2 \geq 0$ (since the buyer acquires no private information, this is equivalent to giving him an ex ante expected utility of $\bar{u}_2$), and satisfying the seller's incentive compatibility constraints.

We now proceed to characterize the solution to problem (23.F.17) through a series of steps.

Step 1: Any solution to problem (23.F.17) has $y_L = 1$; that is, in any interim incentive efficient social choice function, trade is certain to occur when the good is of low quality.

To see this, suppose that $(y_L^*, t_L^*, y_H^*, t_H^*)$ solves (23.F.17) but that $y_L^* < 1$. Consider a change to social choice function $(\hat{y}_L, \hat{t}_L, \hat{y}_H, \hat{t}_H) = (y_L^* + \varepsilon, t_L^* + 30\varepsilon, y_H^*, t_H^*)$

53. In essence, we have expressed all of these in terms of the agents' *gains from trade*.

where $\varepsilon > 0$. For a sufficiently small $\varepsilon > 0$, this new social choice function satisfies all of the constraints of problem (23.F.17) (check this), and raises the value of the objective function—but this contradicts the optimality of $(y_L^*, t_L^*, y_H^*, t_H^*)$.

Step 2: Any solution to problem (23.F.17) has $y_H < 1$; that is, in any interim incentive efficient social choice function, trade does not occur with certainty when the good is of high quality.

Given step 1, if a solution to (23.F.17), say $(y_L^*, t_L^*, y_H^*, t_H^*)$ has $y_H^* = 1$, then $(y_L^*, t_L^*, y_H^*, t_H^*)$ is ex post (classically) efficient (i.e., it has $y_L^* = y_H^* = 1$). But we have already noted above that no such social choice function is incentive feasible (i.e., is an element of F^*).

Step 3: In any solution to (23.F.17), constraint (ii) is binding (i.e., holds with equality).

Suppose that the social choice function $(y_L^*, t_L^*, y_H^*, t_H^*)$ is a solution to (23.F.17) in which constraint (ii) is not binding in the solution. Consider instead the social choice function $(\hat{y}_L, \hat{t}_L, \hat{y}_H, \hat{t}_H) = (y_L^*, t_L^* + \varepsilon, y_H^* + \varepsilon, t_H^* + 45\varepsilon)$ for $\varepsilon > 0$. For small enough $\varepsilon > 0$, this alternative social choice function satisfies all of the constraints of problem (23.F.17) (note that it satisfies $\hat{y}_H < 1$ because, by step 2, $y_H^* < 1$; check the other constraints too). Moreover, it yields a larger value of the objective function of (23.F.17) than $(y_L^*, t_L^*, y_H^*, t_H^*)$—a contradiction. This establishes step 3.

Step 4: If constraint (ii) binds and $y_L \geq y_H$, then constraint (i) is necessarily satisfied.

If constraint (ii) binds then $t_H - t_L = 20(y_H - y_L)$. If $y_L \geq y_H$ then this implies that $t_H - t_L \geq 40(y_H - y_L)$, or, equivalently, $t_H - 40y_H \geq t_L - 40y_L$. Hence, constraint (i) is satisfied.

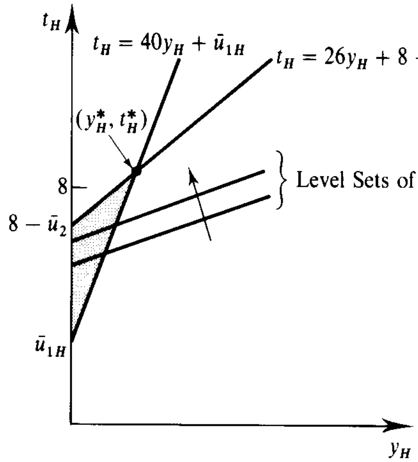
Given steps 1 to 4, we can simplify problem (23.F.17). In particular, we see that (y_L, t_L, y_H, t_H) is interim incentive efficient if and only if $y_L = 1$ and (t_L, y_H, t_H) solve

$$\begin{aligned} \underset{(t_L \in [0,1],\, y_H,\, t_H \in [0,1])}{\text{Max}} \quad & t_L - 20 \\ \text{s.t. (ii}') \quad & t_L - 20 = t_H - 20y_H. \\ \text{(iii)} \quad & t_H - 40y_H \geq \bar{u}_{1H}. \\ \text{(iv)} \quad & .2(50y_H - t_H) + .8(30 - t_L) \geq \bar{u}_2. \end{aligned} \tag{23.F.18}$$

Substituting from constraint (ii′) of problem (23.F.18) for t_L in its objective function and in constraint (iv) we see that we can determine the optimal values of (y_H, t_H) by solving

$$\begin{aligned} \underset{(y_H \in [0,1],\, t_H)}{\text{Max}} \quad & t_H - 20y_H \\ \text{s.t.} \quad \text{(ii)} \quad & t_H - 40y_H \geq \bar{u}_{1H}. \\ \text{(iv}') \quad & 26y_H - t_H + 8 \geq \bar{u}_2. \end{aligned} \tag{23.F.19}$$

The solution for a given pair of values of $\bar{u}_{1H} \geq 0$ and $\bar{u}_2 \geq 0$ is depicted in Figure 23.F.1. The pairs (y_H, t_H) that satisfy constraints (ii) and (iv′) of (23.F.19) lie in the shaded set. Also drawn are two level sets of the objective function $t_H - 20y_H$. The optimal pair (y_H, t_H) for these values of $\bar{u}_{1H}$ and $\bar{u}_2$ is the point labeled (y_H^*, t_H^*). The corresponding values of t_L and y_L in this interim incentive efficient social choice function are then $y_L^* = 1$ and $t_L^* = 20 + t_H^* - 20y_H^*$.



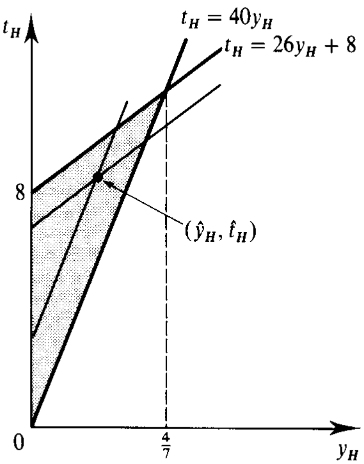


Figure 23.F.1 (left)
The optimal level of (y_H, t_H) in problem (23.F.19) for a given pair $(\bar{u}_H, \bar{u}_2) \geq 0$.

Figure 23.F.2 (right)
The shaded set contains those pairs (y_H, t_H) arising in interim incentive efficient social choice functions.

The shaded set in Figure 23.F.2, $\{(y_H, t_H)\colon y_H \in [0, 1],\ t_H \geq 40y_H, \text{ and } t_H \leq 26y_H + 8\}$, depicts *all* of the pairs (y_H, t_H) that arise in interim incentive efficient social choice functions. These are determined by performing the analysis in Figure 23.F.1 for each possible pair $(\bar{u}_{1H}, \bar{u}_2) \geq 0$ [a sample pair $(\hat{y}_H, \hat{t}_H)$ is also depicted in Figure 23.F.2]. Note that in any interim incentive efficient social choice function we have $y_H \leq 4/7$. As above, for each pair $(\hat{y}_H, \hat{t}_H)$ in this set, we can determine an interim incentive efficient social choice function $(\hat{y}_L, \hat{t}_L, \hat{y}_H, \hat{t}_H)$ by setting $\hat{y}_L = 1$ and $\hat{t}_L = 20 + \hat{t}_H - 20\hat{y}_H$.

Now, out of this set of interim incentive efficient social choice functions, which are ex ante incentive efficient? (Recall that the set of ex ante incentive efficient social choice functions is a subset of the interim incentive efficient set. Note also that although we now employ an ex ante welfare criterion, the set of participation constraints defining F^* continue to be the same interim participation constraints used above.) The buyer's and seller's ex ante expected utilities in an interim incentive efficient social choice function (y_L, t_L, y_H, t_H) are

$$U_1 = .8(t_L + 20(1 - y_L)) + .2(t_H + 40(1 - y_H))$$

and

$$U_2 = .8(30y_L - t_L) + .2(50y_H - t_H).$$

Since $y_L = 1$ and $t_L = 20 + t_H - 20y_H$ in any interim incentive efficient social choice function, these expected utilities can be written as functions of only (y_H, t_H) as follows:

$$U_1 = 24 + t_H - 24y_H, \qquad U_2 = 8 + 26y_H - t_H.$$

In Figure 23.F.3, for an arbitrary point $(\hat{y}_H, \hat{t}_H)$ in set $\{(y_H, t_H)\colon y_H \in [0, 1], t_H \geq 40y_H, \text{ and } t_H \leq 26y_H + 8\}$, the pairs (y_H, t_H) in the shaded set raise the ex ante expected utilities of both the buyer and the seller. As can be seen from this figure, no pair (y_H, t_H) that is not on the $t_H = 40y_H$ boundary of the set $\{(y_H, t_H)\colon y_H \in [0, 1], t_H \geq 40y_H, \text{ and } t_H \leq 26y_H + 8\}$ can be ex ante incentive efficient. Moreover, each pair (y_H, t_H) on this boundary *is* part of an ex ante incentive efficient social choice function. The set of (y_H, t_H) pairs in ex ante incentive efficient social choice functions

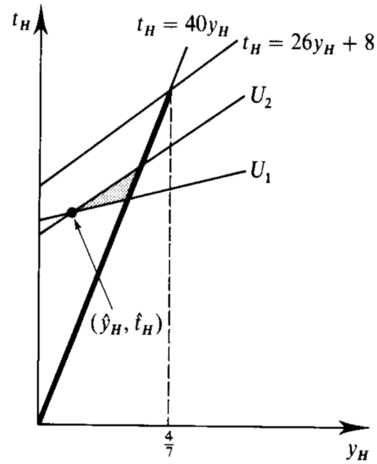


Figure 23.F.3
The set of ex ante incentive efficient social choice functions corresponds to those interim incentive efficient social choice functions with (y_H, t_H) lying in the heavily traced line segment.

is therefore precisely the heavily traced line segment in Figure 23.F.3.[54] Notice that in every such social choice function the interim individual rationality constraint for a high-quality seller binds: the high-quality seller receives no gains from trade. ■

APPENDIX A: IMPLEMENTATION AND MULTIPLE EQUILIBRIA

The notion of implementation that we have employed throughout the chapter (e.g., in Definition 23.B.4) is weaker in one potentially important respect than what we might want: Although a mechanism Γ may implement the social choice function $f(\cdot)$ in the sense of having *an* equilibrium whose outcomes coincide with $f(\cdot)$ for all $\theta \in \Theta$, there may be *other* equilibria of Γ whose outcomes do *not* coincide with $f(\cdot)$. In essence, we have implicitly assumed that the agents will play the equilibrium that the mechanism designer wants if there is more than one.[55]

This suggests that if a mechanism designer wishes to be fully confident that the mechanism Γ does indeed yield the outcomes associated with $f(\cdot)$, he might instead want to insist upon the stronger notion of implementation given in Definition 23.AA.1 (as in Definition 23.B.4, we are deliberately vague here about the equilibrium concept to be employed).

Definition 23.AA.1: The mechanism $\Gamma = (S_1, \ldots, S_I, g(\cdot))$ *strongly implements* social choice function $f: \Theta_1 \times \cdots \times \Theta_I \to X$ if every equilibrium strategy

54. Note that $y_H \leq 4/7 < 1$ in any ex ante incentive efficient social choice function. This may seem at odds with our conclusion in Section 13.B that the ex ante efficient outcome that gives firms zero expected profits has all workers accepting employment in a firm (the structure of the model in Section 13.B parallels that here). The difference is that in Section 13.B we did not impose any interim individual rationality constraints on the workers, effectively supposing that the government could compel workers to participate (pay any taxes, etc.). See Exercise 23.F.10.

55. One possible argument for this assumption is that in a direct revelation mechanism that truthfully implements social choice function $f(\cdot)$, the truth-telling equilibrium may be focal (in the sense discussed in Section 8.D).

profile $(s_1^*(\cdot), \ldots, s_I^*(\cdot))$ of the game induced by Γ has the property that $g(s_1^*(\theta_1), \ldots, s_I^*(\theta_I)) = f(\theta_1, \ldots, \theta_I)$ for all $(\theta_1, \ldots, \theta_I)$.[56]

Let us consider first the implications of this stronger concept for implementation in dominant strategy equilibria. In Exercise 23.C.8 we have already seen a case where, in the direct revelation mechanism that truthfully implements a social choice function $f(\cdot)$ in dominant strategies, some player has more than one dominant strategy, and when he plays one of these dominant strategies the outcome in $f(\cdot)$ does not result (see also Exercises 23.AA.1 and 23.AA.2). Thus, with dominant strategy implementation we may have mechanisms that implement a social choice function $f(\cdot)$ but that do not strongly implement it.

Nevertheless, there are at least two reasons why the multiple equilibrium problem may not be too severe with dominant strategy implementation. First, whenever each agent's dominant strategy in a mechanism Γ that implements $f(\cdot)$ is in fact a *strictly* dominant strategy (rather than just a weakly dominant one), mechanism Γ also strongly implements $f(\cdot)$. This is always the case, for example, in any environment in which agents' preferences never involve indifference between any two elements of X. Second, when a player has two weakly dominant strategies, he is of necessity indifferent between them for any strategies than the other agents choose. Thus, to play the "right" equilibrium in this case, it is only necessary that each agent be willing to resolve his indifference in the way we desire.

In contrast, with Nash-based equilibrium concepts such as Bayesian Nash equilibrium, if mechanism Γ has two equilibria, then in each equilibrium each player may have a strict preference for his equilibrium strategy given that the other agents are playing their respective equilibrium strategies. Having agents play the "right" equilibrium is then not just a matter of resolving indifference but rather of generating expectations that the desired equilibrium is the one that will occur. Example 23.AA.1 illustrates the problem.

Example 23.AA.1: *Multiple Equilibria in the Expected Externality Mechanism.* Consider again the expected externality mechanism of Section 23.D. Suppose that we are in a setting with two agents ($I = 2$) in which a decision must be made regarding a public project (see Example 23.B.3). The project may be either done ($k = 1$) or not done ($k = 0$). Each agent's valuation (net of funding the project) is either θ_L or θ_H (so $\Theta_i = \{\theta_L, \theta_H\}$ for $i = 1, 2$), where $\theta_H > 0 > \theta_L$ and $\theta_L + \theta_H > 0$. The agents' valuations are statistically independent with Prob $(\theta_i = \theta_L) = \lambda \in (0, 1)$ for $i = 1, 2$.

In the expected externality mechanism, each agent i announces his valuation and agent i's transfer when the announced types are (θ_1, θ_2) has the form $t_i(\theta_i, \theta_{-i}) = E_{\tilde{\theta}_{-i}}[\tilde{\theta}_{-i} k^*(\theta_i, \tilde{\theta}_{-i})] + h_i(\theta_{-i})$, where $k^*(\theta_1, \theta_2) = 0$ if $\theta_1 = \theta_2 = \theta_L$, and $k^*(\theta_1, \theta_2) = 1$ otherwise.

As we saw in Section 23.D, in one Bayesian Nash equilibrium of this mechanism, truth telling is each agent's equilibrium strategy. But this truth-telling equilibrium is *not* the only Bayesian Nash equilibrium. In particular, there is an equilibrium in which both agents always claim that θ_H is their type. To see this, consider agent i's optimal

56. The "strong" terminology is not standard; in the literature it is not uncommon, for example, to see the strong implementation concept simply referred to as "implementation."

strategy if agent $-i$ will always announce θ_H. Whichever announcement agent i makes, the project is done. Thus, regardless of his actual type, agent i's direct benefit (i.e., $\theta_i k^*(\theta_1, \theta_2)$) is not affected by his announcement (it is θ_L if he is of type θ_L, and θ_H if he is of type θ_H). It follows that agent i's optimal strategy is to make an announcement that maximizes his expected transfer. Now, agent i's expected transfer if he announces θ_H is $(\lambda\theta_L + (1-\lambda)\theta_H) + h_i(\theta_H)$, whereas if he announces θ_L his expected transfer is $(1-\lambda)\theta_H + h_i(\theta_H)$. Hence, agent i will prefer to announce θ_H regardless of his type if agent $-i$ is doing the same. It follows that both agents always announcing θ_H, and the project consequently always being done, constitutes a second Bayesian Nash equilibrium of this mechanism. ■

We shall not pursue here the characterization of social choice functions that can be strongly implemented in Bayesian Nash equilibria. A good source of further reading on this subject is Palfrey (1992). We also refer to Appendix B, where we discuss many of these issues for the special context of complete information environments.

There are, however, two important points about strong implementation that we wish to stress here. First, when trying to strongly implement a social choice function $f(\cdot)$, we *cannot* generally restrict attention to direct revelation mechanisms. The reason is that when we replace a mechanism $\Gamma = (S_1, \ldots, S_I, g(\cdot))$ with a direct revelation mechanism, as envisioned by the revelation principle, we may introduce new, undesirable equilibria. (See Exercises 23.C.8 and 23.AA.1 for an illustration.) Second, because a social choice function $f(\cdot)$ can be strongly implemented only if it can be implemented in the weaker sense studied in the text of the chapter, all of the necessary conditions for implementation that we have derived are *still* necessary for $f(\cdot)$ to be strongly implemented. Thus, for example, the conclusions of the Gibbard–Satterthwaite theorem (Proposition 23.C.3), the revenue equivalence theorem (Proposition 23.D.3), and the Myerson–Satterthwaite theorem (Proposition 23.E.1) all continue to be valid when we seek strong implementation.

Throughout the chapter we have restricted attention to single-valued social choice functions. It is sometimes natural, however, to consider social choice *correspondences* that can specify more than one acceptable alternative for a given profile of agent types. In this case, we would say that mechanism $\Gamma = (S_1, \ldots, S_I, g(\cdot))$ strongly implements the social choice correspondence $f(\cdot)$ if every equilibrium $s^*(\cdot)$ of the game induced by Γ has the property that $g(s^*(\theta)) \in f(\theta)$, that is, if, for every θ, all possible equilibrium outcomes are acceptable alternatives according to $f(\cdot)$.

APPENDIX B: IMPLEMENTATION IN ENVIRONMENTS WITH COMPLETE INFORMATION

In this appendix we provide a brief discussion of implementation in complete information environments. An excellent source for further reading on this subject is the survey by Moore (1992) [see also Maskin (1985)].

In the complete information case we suppose that each agent will observe not only his own preference parameter θ_i, but also the preference parameters θ_{-i} of all other agents. However, while the agents will observe each others' preference parameters, we suppose that no outsider does. Thus, despite the agents' abilities to observe θ, there is still an implementation problem: Because no outsider (such as a court) will observe θ, the agents cannot write an enforceable ex ante agreement saying that they will choose outcome $f(\theta)$ when agents' preferences are θ. Rather, they can only agree to participate in some mechanism in which equilibrium play yields $f(\theta)$ if θ is realized.[57]

Note that a complete information setting can be viewed as a special case of the general environment considered throughout this chapter, in which the probability density $\phi(\cdot)$ on Θ has the (degenerate) property that each agent's observation of his "type" is completely informative about the types of the other agents.[58]

To begin, observe that the set of social choice functions that are dominant strategy implementable is unaffected by the presence of complete information because an agent's belief about the types of other agents does not affect his set of dominant strategies.[59] [Indeed, recall our comment in Section 23.C that if mechanism Γ implements social choice function $f(\cdot)$ in dominant strategies, then it does so for *any* $\phi(\cdot)$.]

This is *not* the case, however, for implementation in Nash-based equilibrium concepts, such as Bayesian Nash equilibrium. Recall that in complete information environments, the Bayesian Nash equilibrium concept reduces to our standard notion of Nash equilibrium. This motivates Definition 23.BB.1.

Definition 23.BB.1: The mechanism $\Gamma = (S_1, \ldots, S_I, g(\cdot))$ *implements the social choice function $f(\cdot)$ in Nash equilibrium* if, for each profile of the agents' preference parameters $\theta = (\theta_1, \ldots, \theta_I) \in \Theta$, there is a Nash equilibrium of the game induced by Γ, $s^*(\theta) = (s_1^*(\theta), \ldots, s_I^*(\theta))$, such that $g(s^*(\theta)) = f(\theta)$. The mechanism $\Gamma = (S_1, \ldots, S_I, g(\cdot))$ *strongly implements the social choice function $f(\cdot)$ in Nash equilibrium* if, for each profile of the agents' preference parameters $\theta = (\theta_1, \ldots, \theta_I) \in \Theta$, every Nash equilibrium of the game induced by Γ results in outcome $f(\theta)$.

The first point to note about implementation in Nash equilibria is that if we are satisfied with the weaker notion of implementation that we have employed throughout the text of the chapter (as opposed to the strong implementation concept discussed in Appendix A), then *any* social choice function can be implemented in Nash equilibrium as long as $I \geq 3$. To see this, consider the following mechanism: each agent i simultaneously announces a profile of types for each of the I agents. If at least $I - 1$ agents announce the same profile, say $\hat{\theta}$, then we select outcome $f(\hat{\theta})$.

57. This type of setting is often natural in contracting problems, where it is frequently reasonable to suppose that the parties will come to know a lot about each other that is not verifiable by any outside enforcer of their contract.

58. Thus, we can think of the complete information environment as a case in which agents receive signals that are perfectly correlated. There are several ways to formalize this. Perhaps the simplest is to suppose that each agent i's preference parameter is drawn from some set Θ_i. An agent's *signal* (or *type*), which is now represented by $\bar{\theta}_i = (\theta_{i1}, \ldots, \theta_{iI}) \in \Theta$, is a vector giving agent i's observation of his and every other agent's preference parameters. Thus, the set of possible "types" for agent i in the sense in which we have used this term throughout the chapter, is now $\bar{\Theta}_i = \Theta$ for each $i = 1, \ldots, I$. The probability density $\phi(\cdot)$ on the set of possible types $\bar{\Theta}_1 \times \cdots \times \bar{\Theta}_I$ then satisfies the property that $\phi(\bar{\theta}_1, \ldots, \bar{\theta}_I) > 0$ if and only if $\bar{\theta}_1 = \cdots = \bar{\theta}_I$, and agent i's Bernoulli utility function has the form $u_i(x, \bar{\theta}_i) = \tilde{u}_i(x, \theta_{ii})$.

59. The same is true for strong implementation in dominant strategies.

Otherwise we select outcome $x_0 \in X$ (x_0 is arbitary). With this mechanism, for every profile θ, there is a Nash equilibrium in which every agent announces θ, and the resulting outcome is $f(\theta)$, because no agent can affect the outcome by unilaterally deviating.

Although this mechanism implements $f(\cdot)$ in Nash equilibrium, it is obviously not a very attractive mechanism [i.e., its implementation of $f(\cdot)$ is not very convincing] because there are so many other Nash equilibria that do not result in outcome $f(\theta)$ when the preference profile is θ. Indeed, with this mechanism, given the profile of preference parameters θ, there is a Nash equilibrium resulting in x for every $x \in f(\Theta) = \{x \in X\colon \text{there is a } \theta \in \Theta \text{ such that } f(\theta) = x\}$. We see then that for Nash implementation with $I \geq 3$, the *entire* problem of satisfactorily implementing a given social choice function revolves around the issue of successfully dealing with the multiple equilibrium problem discussed in Appendix A.

Given this observation, what social choice functions can we strongly implement in Nash equilibrium? The simple but powerful result in Proposition 23.BB.1 comes from Maskin's (1977) path-breaking paper on Nash implementation.

Proposition 23.BB.1: If the social choice function $f(\cdot)$ can be strongly implemented in Nash equilibrium, then $f(\cdot)$ is monotonic.[60]

Proof: Suppose that $\Gamma = (S_1, \ldots, S_I, g(\cdot))$ strongly implements $f(\cdot)$. Then when the preference parameter profile is θ, there is a Nash equilibrium resulting in outcome $f(\theta)$; that is, there is a strategy profile $s^* = (s_1^*, \ldots, s_I^*)$ having the properties that $g(s^*) = f(\theta)$ and $g(\hat{s}_i, s_{-i}^*) \in L_i(f(\theta), \theta_i)$ for all $\hat{s}_i \in S_i$ and all i.[61] Now suppose that $f(\cdot)$ is not monotonic. Then there exists another profile of preference parameters $\theta' \in \Theta$ such that $L_i(f(\theta), \theta_i) \subset L_i(f(\theta), \theta_i')$ for all i, but $f(\theta') \neq f(\theta)$. But s^* is also a Nash equilibrium under preference parameter profile θ', because $g(\hat{s}_i, s_{-i}^*) \in L_i(f(\theta), \theta_i')$ for all $\hat{s}_i \in S_i$ and all i. Hence, Γ does not strongly implement $f(\cdot)$—a contradiction. ■

As an illustration of the restriction imposed by monotonicity, Proposition 23.BB.2 records one implication of this result.

Proposition 23.BB.2: Suppose that X is finite and contains at least three elements, that $\mathscr{R}_i = \mathscr{P}$ for all i, and that $f(\Theta) = X$. Then the social choice function $f(\cdot)$ is strongly implementable in Nash equilibrium if and only if it is dictatorial.

Proof: To strongly implement a dictatorial social choice function in Nash equilibrium, we need only let the dictator choose an alternative from X. In the other direction, the result follows from steps 2 and 3 of the proof of the Gibbard–Satterthwaite theorem (Proposition 23.C.3). ■

One lesson to be learned from Propositions 23.BB.1 and 23.BB.2 is that dealing with the multiple equilibrium problem can potentially impose very significant restrictions on the set of implementable social choice functions.

60. See Definition 23.C.5.

61. Recall that $L_i(x, \theta_i) \subset X$ is agent i's lower contour set for outcome $x \in X$ when agent i's preference parameter is θ_i.

Maskin (1977) also showed that, when $I > 2$, monotonicity is *almost*, but not quite, a sufficient condition for strong implementation [we omit the proof; see Moore (1992) for a discussion of this and more recent results, including consideration of the case $I = 2$]. Maskin's added condition, known as *no veto power*, requires that if $I - 1$ agents all rank some alternative x as their best alternative then $x = f(\theta)$.

Proposition 23.BB.3: If $I \geq 3$, $f(\cdot)$ is monotonic, and $f(\cdot)$ satisfies no veto power, then $f(\cdot)$ is strongly implementable in Nash equilibrium.

No veto power should be thought of as a very weak additional requirement; indeed, in any setting in which there is a desirable and transferable private good, it is trivially satisfied: no two agents then ever have the same top-ranked alternative (each wants to get all of the available transferable private good). Thus, in these commonly studied environments, monotonicity of $f(\cdot)$ is a necessary and sufficient condition for $f(\cdot)$ to be strongly implementable in Nash equilibrium.

A multivalued social choice correspondence $f(\cdot)$ is said to be monotonic if whenever $x \in f(\theta)$ and $L_i(x, \theta_i) \subset L_i(x, \theta_i')$ for all i then $x \in f(\theta')$. The necessary and sufficient conditions for Nash implementation in Propositions 23.BB.1 and 23.BB.3 carry over to the multivalued case. [In fact, for Proposition 23.BB.3, Maskin's result actually establishes that there is a mechanism that has, for each preference profile θ, a set of Nash equilibrium outcomes exactly equal to the set $f(\theta)$; this type of implementation of a social choice correspondence is commonly called *full implementation.*] It may be verified, for example, that if $f(\theta)$ is equal to the Pareto set for all θ (the set of all ex post efficient outcomes in X given preference profile θ), then $f(\cdot)$ is monotonic. Thus, in any setting with a transferable good, the Pareto set social choice correspondence is strongly implementable in Nash equilibrium.

Implementation using Extensive Form Games: Subgame Perfect Implementation

So far we have seen that the need to "knock out" undesirable equilibria (formalized through the notion of strong implementation) can significantly restrict the set of implementable social choice functions. This suggests the possibility that we may be more successful if we use instead a refinement of the Nash equilibrium concept. Indeed, recent work has shown that such refinements can be very powerful. Here we briefly illustrate how, by considering *dynamic* mechanisms and employing the equilibrium concepts for dynamic games discussed in Section 9.B, we can expand the set of strongly implementable social choice functions.

Example 23.BB.1: [Adapted from Moore and Repullo (1988)]. Consider a pure exchange economy (see Example 23.B.2) with two consumers in which each consumer has two possible individualistic preference relations: if $\theta_i = \theta_i^C$, then consumer i has Cobb–Douglas preferences; if $\theta_i = \theta_i^L$, then consumer i has Leontief preferences. These two possible preference relations for consumers 1 and 2 are depicted in Figure 23.BB.1.

Suppose that we wish to strongly implement the social choice function

$$f(\theta) = \begin{cases} x^C & \text{if } \theta_1 = \theta_1^C, \\ x^L & \text{if } \theta_1 = \theta_1^L, \end{cases}$$

where x^C and x^L are the allocations depicted in Figure 23.BB.1. Note that consumer 1 always prefers x^C to x^L, and the reverse is true for consumer 2. By inspection of

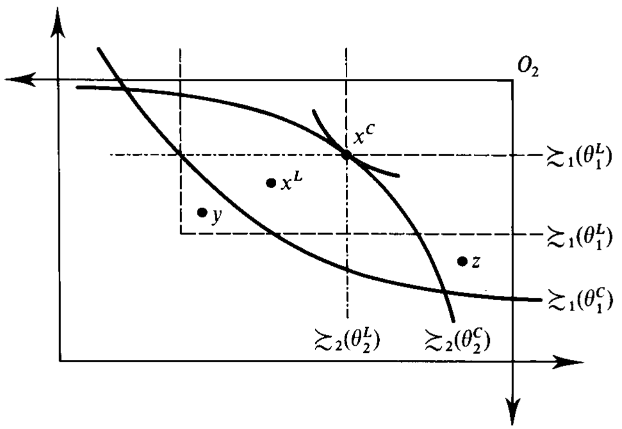


Figure 23.BB.1 Preferences and outcomes in Example 23.BB.1.

Figure 23.BB.1, we see that $f(\cdot)$ is not monotonic, because $L_i(x^C, \theta_i^C) \subset L_i(x^C, \theta_i^L)$ for $i = 1, 2$ but $f(\theta_1^L, \theta_2^L) \neq f(\theta_1^C, \theta_2^C)$. Hence, by Proposition 23.BB.1, $f(\cdot)$ cannot be strongly implemented in Nash equilibrium.

Suppose, instead, that we construct the following three-stage dynamic mechanism:

Stage 1: Agent 1 announces either "L" or "C." If he announces "L," x^L is immediately chosen. If he announces "C," we go to stage 2.

Stage 2: Agent 2 says either "agree" or "challenge." If he says "agree," then x^C is immediately chosen. If he says "challenge," then we go to stage 3.

Stage 3: Agent 1 chooses between the allocations y and z depicted in Figure 23.BB.1.

It is straightforward to verify that, for each possible profile of preferences $\theta = (\theta_1, \theta_2)$, the unique subgame perfect Nash equilibrium of this dynamic game of perfect information results in outcome $f(\theta)$ (see Exercise 23.BB.1). Thus, $f(\cdot)$ *can* be strongly implemented if we consider dynamic mechanisms and take subgame perfect Nash equilibrium as the appropriate solution concept for the games induced by these mechanisms. ■

In fact, Moore and Repullo (1988) [see also Moore (1990)] show that the use of dynamic mechanisms and subgame perfection expands the set of strongly implementable social choice functions dramatically compared to the use of the Nash equilibrium concept. Even stronger results are possible with other refinements; see, for example, Palfrey and Srivastava (1991) for a study of strong implementation in undominated Nash equilibrium (i.e., Nash equilibria in which no agent is playing a weakly dominated strategy).[62]

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EXERCISES

23.B.1$^{\mathrm{A}}$ Consider the setting explored in Example 23.B.1, where $\mathscr{R}_1 = \{\succsim_1(\bar{\theta}_1)\}$ and $\mathscr{R}_2 = \{\succsim_2(\theta_2'), \succsim_2(\theta_2'')\}$. For each of the following social choice functions $f(\cdot)$, will agent 2 be willing to truthfully reveal his preferences?

(a) $f(\bar{\theta}_1, \theta_2') = y$, $f(\bar{\theta}_1, \theta_2'') = y$.

(b) $f(\bar{\theta}_1, \theta_2') = z$, $f(\bar{\theta}_1, \theta_2'') = x$.

(c) $f(\bar{\theta}_1, \theta_2') = z$, $f(\bar{\theta}_1, \theta_2'') = y$.

(d) $f(\bar{\theta}_1, \theta_2') = z$, $f(\bar{\theta}_1, \theta_2'') = z$.

(e) $f(\bar{\theta}_1, \theta_2') = y$, $f(\bar{\theta}_1, \theta_2'') = z$.

23.B.2$^{\mathrm{A}}$ Consider a bilateral trade setting (see Example 23.B.4) in which both the seller's (agent 1) and the buyer's (agent 2) types are drawn independently from the uniform distribution on $[0, 1]$. Suppose that we try to implement the social choice function $f(\cdot) = (y_1(\cdot), y_2(\cdot), t_1(\cdot), t_2(\cdot))$ such that

$$y_1(\theta_1, \theta_2) = 1 \text{ if } \theta_1 \geq \theta_2; \; = 0 \text{ if } \theta_1 < \theta_2.$$

$$y_2(\theta_1, \theta_2) = 1 \text{ if } \theta_2 > \theta_1; \; = 0 \text{ if } \theta_2 \leq \theta_1.$$

$$t_1(\theta_1, \theta_2) = \tfrac{1}{2}(\theta_1 + \theta_2) y_2(\theta_1, \theta_2).$$

$$t_2(\theta_1, \theta_2) = -\tfrac{1}{2}(\theta_1 + \theta_2) y_2(\theta_1, \theta_2).$$

Suppose that the seller truthfully reveals his type for all $\theta_1 \in [0, 1]$. Will the buyer find it worthwhile to reveal his type? Interpret.

23.B.3$^{\mathrm{B}}$ Show that $b_i(\theta_i) = \theta_i$ for all $\theta_i \in [0, 1]$ is a weakly dominant strategy for each agent i in the second-price sealed-bid auction.

23.B.4$^{\mathrm{C}}$ Consider a bilateral trade setting (see Example 23.B.4) in which both the seller's and the buyer's types are drawn independently from the uniform distribution on $[0, 1]$.

(a) Consider the *double auction* mechanism in which the seller (agent 1) and buyer (agent 2) each submit a sealed bid, $b_i \geq 0$. If $b_1 \geq b_2$, the seller keeps the good and no monetary transfer is made; while if $b_2 > b_1$, the buyer gets the good and pays the seller the amount $\frac{1}{2}(b_1 + b_2)$. (The interpretation is that the seller's bid is his minimum acceptable price, while the buyer's is his maximum acceptable price; if trade occurs, the price splits the difference between these amounts.) Solve for a Bayesian Nash equilibrium of this game in which each agent i's strategy takes the form $b_i(\theta_i) = \alpha_i + \beta_i \theta_i$. What social choice function does this equilibrium of this mechanism implement? Is it ex post efficient?

(b) Show that the social choice function derived in **(a)** is incentive compatible; that is, that it can be truthfully implemented in Bayesian Nash equilibrium.

23.C.1$^{\mathrm{A}}$ Verify that if the preference reversal property [condition (23.C.6)] is satisfied for all i and all θ_i', θ_i'', and θ_{-i}, then $f(\cdot)$ is truthfully implementable in dominant strategies.

23.C.2$^{\mathrm{B}}$ Show that, for any I, when X contains two elements (say, $X = \{x_1, x_2\}$), then any majority voting social choice function [i.e., a social choice function that always chooses alternative x_i if more agents prefer x_i over x_j than prefer x_j over x_i (it may select either x_1 or x_2 if the number of agents preferring x_1 over x_2 equals the number preferring x_2 over x_1)] is truthfully implementable in dominant strategies.

23.C.3[A] Show that when $\mathscr{R}_i = \mathscr{P}$ for all i, any ex post efficient social choice function $f(\cdot)$ has $f(\Theta) = X$.

23.C.4[A] Show that if $f\colon \Theta \to X$ is truthfully implementable in dominant strategies when the set of possible types is Θ_i for $i = 1, \ldots, I$, then when each agent i's set of possible types is $\hat{\Theta}_i \subset \Theta_i$ (for $i = 1, \ldots, I$) the social choice function $\hat{f}\colon \hat{\Theta} \to X$ satisfying $\hat{f}(\theta) = f(\theta)$ for all $\theta \in \hat{\Theta}$ is truthfully implementable in dominant strategies.

23.C.5[C] Show that in an environment with single-peaked preferences having the property that no two alternatives are indifferent (see Section 21.D) and an odd number of agents, the (unique) social choice function that always selects a Condorcet winner (see Section 21.D) is truthfully implementable in dominant strategies.

23.C.6[C] The property of a social choice function identified in Proposition 23.C.2 is called *independent person-by-person monotonicity* (IPM) by Dasgupta, Hammond, and Maskin (1979). In this exercise, we investigate its relationship with the *monotonicity* property defined in Definition 23.C.5.

(a) Show by means of an example that if $f(\cdot)$ satisfies IPM, it need not be monotonic (this can be done with a very simple example).

(b) Show by means of an example that if $f(\cdot)$ is monotonic, it need not satisfy IPM.

(c) Prove that if $f(\cdot)$ satisfies IPM, and if $\mathscr{R}_i \subset \mathscr{P}$ for all i, then $f(\cdot)$ is monotonic.

(d) Prove that if $f(\cdot)$ is monotonic, and $\mathscr{R}_i = \mathscr{P}$ for all i, then $f(\cdot)$ satisfies IPM.

23.C.7[C] A social welfare functional $F(\cdot)$ (see Section 21.C) satisfies the property of *nonnegative responsiveness* if for all $x, y \in X$, and for any two pairs of profiles of preferences for the I agents $(\succsim_1, \ldots, \succsim_I)$ and $(\succsim'_1, \ldots, \succsim'_I)$ such that $x \succsim_i y \Rightarrow x \succsim'_i y$ and $x \succ_i y \Rightarrow x \succ'_i y$ for all i, we have

$$x\, F(\succsim_1, \ldots, \succsim_I)\, y \Rightarrow x\, F(\succsim'_1, \ldots, \succsim'_I)\, y$$

and

$$x\, F_p(\succsim_1, \ldots, \succsim_I)\, y \Rightarrow x\, F_p(\succsim'_1, \ldots, \succsim'_I)\, y,$$

where $x\, F_p(\cdot)\, y$ means "$x\, F(\cdot)\, y$ and not $y\, F(\cdot)\, x$." Show that if the social choice function $f(\cdot)$ maximizes a social welfare functional $F(\cdot)$ satisfying nonnegative responsiveness [in the sense that for all $(\theta_1, \ldots, \theta_I)$ we have $f(\theta_1, \ldots, \theta_I) = \{x \in X\colon x\, F(\succsim_1(\theta_1), \ldots, \succsim_I(\theta_I))\, y \text{ for all } y \in X\}$], then $f(\cdot)$ is truthfully implementable in dominant strategies.

23.C.8[A] Suppose that $I = 2$, $X = \{a, b, c, d, e\}$, $\Theta_1 = \{\theta'_1, \theta''_1\}$, and $\Theta_2 = \{\theta'_2, \theta''_2\}$, and that the agents' possible preferences are (a–b means that alternatives a and b are indifferent):

$\succsim_1(\theta'_1)$	$\succsim_1(\theta''_1)$	$\succsim_2(\theta'_2)$	$\succsim_2(\theta''_2)$
a–b	a	a–b	a
c	b	c	b
d	d	d	d
e	c	e	c
	e		e

Consider the social choice function

$$f(\theta) = \begin{cases} b & \text{if } \theta = (\theta_1', \theta_2'), \\ a & \text{otherwise.} \end{cases}$$

(a) Is $f(\cdot)$ ex post efficient?

(b) Does it satisfy the property identified in Proposition 23.C.2?

(c) Examine the direct revelation mechanism that truthfully implements $f(\cdot)$. Is truth telling each agent's *unique* (weakly) dominant strategy? Show that if an agent chooses his untruthful (weakly) dominant strategy, then $f(\cdot)$ is not implemented.

23.C.9^{C} Suppose that $K = \mathbb{R}$, the $v_i(\cdot, \theta_i)$ functions are assumed to be twice continuously differentiable, θ_i is drawn from an interval $[\underline{\theta}_i, \bar{\theta}_i]$, $\partial^2 v_i(k, \theta_i)/\partial k^2 < 0$, and $\partial^2 v_i(k, \theta_i)/\partial k\, \partial\theta_i > 0$. Show that the continuously differentiable social choice function $f(\cdot) = (k(\cdot), t_1(\cdot), \ldots, t_I(\cdot))$ is truthfully implementable in dominant strategies if and only if, for all $i = 1, \ldots, I$,

$$k(\theta) \text{ is nondecreasing in } \theta_i$$

and

$$t_i(\theta_i, \theta_{-i}) = t_i(\underline{\theta}_i, \theta_{-i}) - \int_{\underline{\theta}_i}^{\theta_i} \frac{\partial v_i(k(s, \theta_{-i}), s)}{\partial k} \frac{\partial k(s, \theta_{-i})}{\partial s}\, ds.$$

23.C.10^{B} (B. Holmstrom) Consider the quasilinear environment studied in Section 23.C. Let $k^*(\cdot)$ denote any project decision rule that satisfies (23.C.7). Also define the function $V^*(\theta) = \sum_i v_i(k^*(\theta), \theta_i)$.

(a) Prove that there exists an ex post efficient social choice function [i.e., one that satisfies condition (23.C.7) *and* the budget balance condition (23.C.12)] that is truthfully implementable in dominant strategies if and only if the function $V^*(\cdot)$ can be written as $V^*(\theta) = \sum_i V_i(\theta_{-i})$ for some functions $V_1(\cdot), \ldots, V_I(\cdot)$ having the property that $V_i(\cdot)$ depends only on θ_{-i} for all i.

(b) Use the result in part **(a)** to show that when $I = 3$, $K = \mathbb{R}$, $\Theta_i = \mathbb{R}_+$ for all i, and $v_i(k, \theta_i) = \theta_i k - (\frac{1}{2})k^2$ for all i an ex post efficient social choice function exists that is truthfully implementable in dominant strategies. (This result extends to any $I > 2$.)

(c) Now suppose that the $v_i(k, \theta_i)$ functions are such that $V^*(\cdot)$ is an I-times continuously differentiable function. Argue that a necessary condition for an ex post efficient social choice function to exist is that, at all θ,

$$\frac{\partial^I V^*(\theta)}{\partial\theta_1 \ldots \partial\theta_I} = 0.$$

(In fact, this is a sufficient condition as well.)

(d) Use the result in **(c)** to verify that, under the assumptions made in the small type discussion at the end of Section 23.C, when $I = 2$ no ex post efficient social choice function is truthfully implementable in dominant strategies.

23.C.11^{A} Consider a quasilinear environment, but now suppose that each agent i has a Bernoulli utility function of the form $u_i(v_i(k, \theta_i) + \bar{m}_i + t_i)$ with $u_i'(\cdot) > 0$. That is, preferences over certain outcomes take a quasilinear form, but risk preferences are unrestricted. Verify that Proposition 23.C.4 is unaffected by this change.

23.D.1^{B} [Based on an example in Myerson (1991)] A buyer and a seller are bargaining over the sale of an indivisible good. The buyer's valuation is $\theta_b = 10$. The seller's valuation takes one of two values: $\theta_s \in \{0, 9\}$. Let t be the period in which trade occurs ($t = 1, 2, \ldots$) and let p be the price agreed. Both the buyer and the seller have discount factor $\delta < 1$.

(a) What is the set X of alternatives in this setting?

(b) Suppose that in a Bayesian Nash equilibrium of this bargaining process, trade occurs immediately when the seller's valuation is 0 and the price agreed to when the seller has valuation θ_s is $(10 + \theta_s)/2$. What is the earliest possible time at which trade can occur when the seller's valuation is 9?

23.D.2[B] Consider a bilateral trade setting in which each θ_i $(i = 1, 2)$ is independently drawn from a uniform distribution on $[0, 1]$.

(a) Compute the transfer functions in the expected externality mechanism.

(b) Verify that truth telling is a Bayesian Nash equilibrium.

23.D.3[A] Reconsider the first-price and second-price sealed-bid auctions studied in Examples 23.B.5 and 23.B.6. Verify that the revenue equivalence theorem holds for the equilibria identified there.

23.D.4[C] Consider a first-price sealed-bid auction with I symmetric buyers. Each buyer's valuation is independently drawn from the interval $[\underline{\theta}, \bar{\theta}]$ according to the strictly positive density $\phi(\cdot)$.

(a) Show that the buyer's equilibrium bid function is nondecreasing in his type.

(b) Argue that in any symmetric equilibrium $(b^*(\cdot), \ldots, b^*(\cdot))$ there can be no interval of types (θ', θ''), $\theta' \neq \theta''$, such that $b^*(\theta)$ is the same for all $\theta \in (\theta', \theta'')$. Conclude that $b^*(\cdot)$ must therefore be strictly increasing.

(c) Argue, using the revenue equivalence theorem, that any symmetric equilibrium of such an auction must yield the seller the same expected revenue as in the (dominant strategy) equilibrium of the second-price sealed-bid auction.

23.D.5[C] For the same assumptions as in Exercise 23.D.4, consider a sealed-bid *all-pay* auction in which every buyer submits a bid, the highest bidder receives the good, and *every* buyer pays the seller the amount of his bid *regardless of whether he wins.* Argue that any symmetric equilibrium of this auction also yields the seller the same expected revenue as the sealed-bid second-price auction. [*Hint*: Follow similar steps as in Exercise 23.D.4.]

23.D.6[C] Suppose that I symmetric individuals wish to acquire the single remaining ticket to a concert. The ticket office opens at 9 a.m. on Monday. Each individual must decide what time to go to get on line: the first individual to get on line will get the ticket. An individual who waits t hours incurs a (monetary equivalent) disutility of βt. Suppose also that an individual showing up after the first one can go home immediately and so incurs no waiting cost. Individual i's value of receiving the ticket is θ_i, and each individual's θ_i is independently drawn from a uniform distribution on $[0, 1]$. What is the expected value of the number of hours that the first individual in line will wait? [*Hint*: Note the analogy to a first-price sealed-bid auction and use the revenue equivalence theorem.] How does this vary when β doubles? When I doubles?

23.E.1[B] Consider again a bilateral trade setting in which each θ_i $(i = 1, 2)$ is independently drawn from a uniform distribution on $[0, 1]$. Suppose now that by refusing to participate in the mechanism a seller with valuation θ_1 receives expected utility θ_1 (he simply consumes the good), whereas a buyer with valuation θ_2 receives expected utility 0 (he simply consumes his endowment of the numeraire, which we have normalized to equal 0). Show that in the expected externality mechanism there is a type of buyer or seller who will strictly prefer not to participate.

23.E.2[A] Argue that when the assumptions of Proposition 23.E.1 hold in the bilateral trade setting:

(a) There is no social choice function $f(\cdot)$ that is dominant strategy incentive compatible and interim individually rational (i.e., that gives each agent i nonnegative gains from participation conditional on his type θ_i, for all θ_i).

(b) There is no social choice function $f(\cdot)$ that is Bayesian incentive compatible and ex post individually rational [i.e., that gives each agent nonnegative gains from participation for every pair of types (θ_1, θ_2)].

23.E.3$^{\mathbf{B}}$ Show by means of an example that when the buyer and seller in a bilateral trade setting both have a discrete set of possible valuations, social choice functions may exist that are Bayesian incentive compatible, ex post efficient, and individually rational. [*Hint*: It suffices to let each have two possible types.] Conclude that the assumption of a strictly positive density is required for the Myerson–Satterthwaite theorem.

23.E.4$^{\mathbf{B}}$ A seller ($i = 1$) and a buyer ($i = 2$) are bargaining over the sale of an indivisible good. Trade can occur at discrete periods $t = 1, 2, \ldots$. Both the buyer and the seller have discount factor $\delta < 1$. The buyer's and seller's valuations are drawn independently with positive densities from $[\underline{\theta}_2, \bar{\theta}_2]$ and $[\underline{\theta}_1, \bar{\theta}_1]$, respectively. Assume that $(\underline{\theta}_2, \bar{\theta}_2) \cap (\underline{\theta}_1, \bar{\theta}_1) \neq \varnothing$. Note that in this setting ex post efficiency requires that trade occur in period 1 whenever $\theta_2 > \theta_1$, and that trade not occur whenever $\theta_1 > \theta_2$. Use the Myerson–Satterthwaite theorem to show that, in this setting with discounting, no voluntary trading process can achieve ex post efficiency.

23.E.5$^{\mathbf{B}}$ Suppose there is a *continuum* of buyers and sellers (with quasilinear preferences). Each seller initially has one unit of an indivisible good and each buyer initially has none. A seller's valuation for consumption of the good is $\theta_1 \in [\underline{\theta}_1, \bar{\theta}_1]$, which is independently and identically drawn from distribution $\Phi_1(\cdot)$ with associated strictly positive density $\phi_1(\cdot)$. A buyer's valuation from consumption of the good is $\theta_2 \in [\underline{\theta}_2, \bar{\theta}_2]$, which is independently and identically drawn from distribution $\Phi_2(\cdot)$ with associated strictly positive density $\phi_2(\cdot)$.

(a) Characterize the trading rule in an ex post efficient social choice function. Which buyers and sellers end up with a unit of the good?

(b) Exhibit a social choice function that has the trading rule you identified in **(a)**, is Bayesian incentive compatible, and is individually rational. [*Hint*: Think of a "competitive" mechanism.] Conclude that the inefficiency identified in the Myerson–Satterthwaite theorem goes away as the number of buyers and sellers grows large. [For a formal examination showing that, with a finite number of traders, the efficiency loss goes to zero as the number of traders grows large, see Gresik and Satterthwaite (1989).]

23.E.6$^{\mathbf{B}}$ Consider a bilateral trading setting in which *both* agents initially own one unit of a good. Each agent i's ($i = 1, 2$) valuation per unit consumed of the good is θ_i. Assume that θ_i is independently drawn from a uniform distribution on $[0, 1]$.

(a) Characterize the trading rule in an ex post efficient social choice function.

(b) Consider the following mechanism: Each agent submits a bid; the highest bidder buys the other agent's unit of the good and pays him the amount of his bid. Derive a symmetric Bayesian Nash equilibrium of this mechanism. [*Hint*: Look for one in which an agent's bid is a linear function of his type.]

(c) What is the social choice function that is implemented by this mechanism? Verify that it is Bayesian incentive compatible. Is it ex post efficient? Is it individually rational [which here requires that $U_i(\theta_i) \geq \theta_i$ for all θ_i and $i = 1, 2$]? Intuitively, why is there a difference from the conclusion of the Myerson–Satterthwaite theorem? [See Cramton, Gibbons, and Klemperer (1987) for a formal analysis of these "partnership division" problems.]

23.E.7$^{\mathbf{B}}$ Consider a bilateral trade setting in which the buyer's and seller's valuations are drawn independently from the uniform distribution on $[0, 1]$.

(a) Show that if $f(\cdot)$ is a Bayesian incentive compatible and interim individually rational social choice function that is ex post efficient, the sum of the buyer's and seller's expected utilities under $f(\cdot)$ cannot be less than 5/6.

(b) Show that, in fact, there is no social choice function (whether Bayesian incentive compatible and interim individually rational or not) in which the sum of the buyer's and seller's expected utilities exceeds 2/3.

23.F.1^C Consider the quasilinear setting studied in Sections 23.C and 23.D. Show that if the social choice function $f(\cdot) \in F^*$ is ex post classically efficient in F_{IR} then it is both ex ante and interim incentive efficient in F^*. [From this fact, we see that if an ex post classically efficient social choice function *can* be implemented in a setting with privately observed types (i.e., if it is incentive feasible), then no other incentive feasible social choice function can welfare dominate it. Note, however, that there may be *other* ex ante or interim incentive efficient social choice functions that are not ex post efficient; for example, you can verify that in Example 23.F.1 there is an ex post classically efficient social choice function that is incentive feasible, but the particular interim incentive efficient social choice function derived in the example is not ex post efficient.]

23.F.2^B [Based on Maskin and Riley (1984)] A monopolist seller produces a good with constant returns to scale at a cost of $c > 0$ per unit. The monopolist sells to a consumer whose preference for the product the monopolist cannot observe. A consumer of type $\theta > 0$ derives a utility of $\theta v(x) - t$ when he consumes x units of the monopolist's product and pays the monopolist a total of t dollars for these units. Assume that $v'(\cdot) > 0$ and $v''(\cdot) < 0$. The set of possible consumer types is $[\underline{\theta}, \bar{\theta}]$ with $\bar{\theta} > \underline{\theta} > 0$, and the distribution of types is $\Phi(\cdot)$, with an associated strictly positive density function $\phi(\cdot) > 0$. Assume that $[\theta - ((1 - \Phi(\theta))/\phi(\theta))]$ is nondecreasing in θ.

Characterize the monopolist's optimal selling mechanism to this consumer, assuming that a consumer of type θ can always choose not to buy at all, thereby deriving a utility of 0.

23.F.3^C An auction with a *reserve price* is an auction in which there is a minimum allowable bid. Suppose that in the auction setting of Example 23.F.2 the I buyers are symmetric and that $\underline{\theta} = 0$. Argue that a second-price sealed-bid auction with a reserve price is an optimal auction in this case. What is the optimal reserve price? Can you think of a modified second-price sealed-bid auction that is optimal in the general (nonsymmetric) case?

23.F.4^B Derive the optimal $y_i(\cdot)$ functions in the auction setting of Example 23.F.2 when the seller's valuation for the object is $\theta_0 > 0$.

23.F.5^B Suppose that a monopolist seller who has two potential buyers has a total of one divisible unit to sell; that is, production costs are zero up to one unit, and infinite beyond that. The demand function of buyer i is the decreasing function $x_i(p)$ for $i = 1, 2$. The monopolist can name distinct prices for the two buyers.

(a) Characterize the monopolist's optimal prices.

(b) Relate your answer in **(a)** to the optimal auction derived in Example 23.F.2. [For more on this, see Bulow and Roberts (1989).]

23.F.6^C [Based on Baron and Myerson (1982)]. Consider the optimal regulatory scheme for a regulator of a monopolist who has known demand function $x(p)$, with $x'(p) < 0$, and a privately observed constant marginal cost of production θ. The regulator can set the monopolist's price and can make a transfer from or to the monopolist, so the set of outcomes is $X = \{(p, t): p > 0 \text{ and } t \in \mathbb{R}\}$. The regulator must guarantee the monopolist a nonnegative profit regardless of his production costs to prevent the monopolist from shutting down. The monopolist's marginal cost θ is drawn from $[\underline{\theta}, \bar{\theta}]$ with $\bar{\theta} > \underline{\theta} > 0$ according to the distribution function $\Phi(\cdot)$, which has an associated strictly positive density function $\phi(\cdot) > 0$. Assume that

$\Phi(\theta)/\phi(\theta)$ is nondecreasing in θ. Denote a type-θ monopolist's profit from outcome (p, t) by $\pi(p, t, \theta) = (p - \theta)x(p) + t$.

(a) Adapt the characterization in Proposition 23.D.2 to this application.

(b) Suppose that the regulator wants to design a direct revelation regulatory scheme $(p(\cdot), t(\cdot))$ that maximizes the expected value of a weighted sum of consumer and producer surplus,

$$\int_{p(\theta)}^{\infty} x(s)\, ds + \alpha\pi(p(\theta), t(\theta), \theta),$$

where $\alpha < 1$. Characterize the regulator's optimal regulatory scheme. What if $\alpha \geq 1$?

23.F.7^{C} [Based on Dana and Spier (1994)] Two firms, $j = 1, 2$, compete for the right to produce in a given market. A social planner designs an optimal auction of production rights to maximize the expected value of social welfare as measured by

$$W = \sum_j \pi_j + S + (\lambda - 1)\sum_j t_j,$$

where t_j denotes the transfer from firm j to the planner, S is consumer surplus, π_j is the gross (pretransfer) profit of firm j, and $\lambda > 1$ is the shadow cost of public funds. The auction specifies transfers for each of the firms and a market structure; that is, it either awards neither firm production rights, awards only one firm production rights (thereby making that firm an unregulated monopolist), or gives production rights to both firms (thereby making them compete as unregulated duopolists).

Each firm j privately observes its fixed cost of production θ_j. The fixed cost levels θ_1 and θ_2 are independently distributed on $[\underline{\theta}, \bar{\theta}]$ with continuously differentiable density function $\phi(\cdot)$ and distribution function $\Phi(\cdot)$. Assume that $\Phi(\cdot)/\phi(\cdot)$ is increasing in θ. The firms have common marginal cost $c < 1$ and produce a homogeneous product for which the market inverse demand function is $p(x) = 1 - x$ (this is publicly known). If both firms are awarded production rights, they interact as Cournot competitors (see Section 12.C).

Characterize the planner's optimal auction of production rights.

23.F.8^{A} Show that any ex post classically efficient social choice function in Example 23.F.3 has $y_L = y_H = 1$.

23.F.9^{B} Show that in the model of Example 23.F.3:

(a) No feasible social choice function is ex post efficient.

(b) In any feasible social choice function, $y_H \leq y_L$ and $t_H \leq t_L$.

(c) In any feasible social choice function, the expected gains from trade of a low-quality seller are at least as large as the expected gains from trade of a high-quality seller; that is, $t_L - 20y_L \geq t_H - 40y_H$.

23.F.10^{B} Characterize the sets of interim and ex ante incentive efficient social choice functions in the model of Example 23.F.3 when trade is not voluntary for the seller (but it is voluntary for the buyer).

23.AA.1^{B} Reconsider Exercise 23.C.8. Exhibit a mechanism $\Gamma = (S_1, \ldots, S_I, g(\cdot))$ that is not a direct revelation mechanism that truthfully implements $f(\cdot)$ in dominant strategies and for which each agent has a *unique* (weakly) dominant strategy.

23.AA.2^{B} Let $K = \{k_0, k_1, \ldots, k_N\}$ be the set of possible projects and suppose that, for each agent i, $\{v_i(\cdot, \theta_i)\colon \theta_i \in \Theta_i\} = \mathscr{V}$, that is, that every possible valuation function from K to $\mathbb{R}$ arises for some $\theta_i \in \Theta_i$. Do players in a Groves mechanism have a unique (weakly) dominant strategy? Consider instead a mechanism in which each agent i is allowed to announce a *normalized* valuation function, that is, a function such that $v_i(k_0) = 0$. Suppose that $k^*(\cdot)$ and the Groves transfers are calculated using these announcements. Does each agent have a unique (weakly) dominant strategy in this normalized Groves mechanism?

23.BB.1^A Consider the dynamic mechanism in Example 23.BB.1.

(a) For each possible preference profile, write down its normal form and identify its Nash equilibria.

(b) For each possible preference profile, identify this mechanism's subgame perfect Nash equilibria.

23.BB.2^B Is a social choice function that is implementable in dominant strategies necessarily implementable in Nash equilibrium? What if we are interested in strong implementation instead?

23.BB.3^C Consider a setting of public project choice (see Example 23.B.3) in which $K = \{0, 1\}$. Let θ_i denote agent i's benefit if the project is done (i.e., if $k = 1$); normalize the value from $k = 0$ to equal zero. Assume that $\Theta_i = \mathbb{R}$. In this setting, the only mechanisms that involve an ex post efficient project choice are Groves mechanisms. Let $k^*(\cdot)$ denote the project choice rule in such a mechanism. Also, suppose that $I \geq 3$. The transfers in a Groves mechanism are characterized by two properties:

(i) if $k^*(\theta_i, \theta_{-i}) = k^*(\theta_i', \theta_{-i})$, then $t_i(\theta_i, \theta_{-i}) = t_i(\theta_i', \theta_{-i})$;
(ii) if $k^*(\theta_i, \theta_{-i}) = 1$ and $k^*(\theta_i', \theta_{-i}) = 0$, then $t_i(\theta_i, \theta_{-i}) - t_i(\theta_i', \theta_{-i}) = \sum_{j \neq i} \theta_j$.

Which, if any, of these two properties must be satisfied by any Nash implementable social choice function that involves an ex post efficient project choice?

Mathematical Appendix

This appendix contains a quick and unsystematic review of some of the mathematical concepts and techniques used in the text.

The formal results are quoted as "Theorems" and they are fairly rigorously stated. It seems useful in a technical appendix such as this to provide motivational remarks, examples, and general ideas for some proofs. This we often do under the label of the "Proof" of the mathematical theorem under discussion. Nonetheless, no rigor of any sort is intended here. Perhaps the heading "Discussion of Theorem" would be more accurate.

It goes without saying that this appendix is no substitute for a more extensive and systematic, book-length, treatment. Good references for some or most of the material covered in this appendix, as well as for further background reading, are Simon and Blume (1993), Sydsaeter and Hammond (1994), Novshek (1993), Dixit (1990), Chang (1984), and Intriligator (1971).

M.A Matrix Notation for Derivatives

We begin by reviewing some matters of notation. The first and most important is that formally and mathematically a "vector" in $\mathbb{R}^N$ is a *column*. This applies to any vector; it does not matter, for example, if the vector represents quantities or prices. It applies also to the *gradient* vector $\nabla f(\bar{x}) \in \mathbb{R}^N$ of a function at a point $\bar{x}$; this is the vector whose nth entry is the partial derivative with respect to the nth variable of the real-valued function $f\colon \mathbb{R}^N \to \mathbb{R}$, evaluated at the point $\bar{x} \in \mathbb{R}^N$. Expositionally, however, because rows take less space to display, we typically describe vectors horizontally in the text, as in $x = (x_1, \ldots, x_N)$. But the rule has no exception: all vectors are mathematically columns.

The inner product of two N vectors $x \in \mathbb{R}^N$ and $y \in \mathbb{R}^N$ is written as $x \cdot y = \sum_n x_n y_n$. If we view these vectors as $N \times 1$ matrices, we see that $x \cdot y = x^{\mathrm{T}} y$, where $^{\mathrm{T}}$ is the matrix transposition operator. An expression such as "$x \cdot$" can always be read as "x^{T}"; for example, the expression $x \cdot A$, where A is an $N \times M$ matrix, is the same as $x^{\mathrm{T}} A$.

If $f\colon \mathbb{R}^N \to \mathbb{R}^M$ is a vector-valued differentiable function, then at any $x \in \mathbb{R}^N$ we denote by $Df(x)$ the $M \times N$ matrix whose mnth entry is $\partial f_m(x)/\partial x_n$. Note, in particular, that if $M = 1$ (so that $f(x) \in \mathbb{R}$) then $Df(x)$ is a $1 \times N$ matrix; in fact $\nabla f(x) = [Df(x)]^{\mathrm{T}}$. To avoid ambiguity, in some cases we write $D_x f(x)$ to indicate explicitly the variables with respect to which the function $f(\cdot)$ is being differentiated. For example, with this notation, if $f\colon \mathbb{R}^{N+K} \to \mathbb{R}^M$ is a function whose arguments are the vectors $x \in \mathbb{R}^N$ and $y \in \mathbb{R}^K$, the matrix $D_x f(x, y)$ is the $M \times N$ matrix whose mnth entry is $\partial f_m(x, y)/\partial x_n$. Finally, for a real-valued differentiable function $f\colon \mathbb{R}^N \to \mathbb{R}$, the *Hessian matrix* $D^2 f(x)$ is the derivative matrix of the vector-valued gradient function $\nabla f(x)$; i.e., $D^2 f(x) = D[\nabla f(x)]$.

In the remainder of this section, we consider differentiable functions and examine how two well-known rules of calculus—the chain rule and the product rule—come out in matrix notation.

The Chain Rule

Suppose that $g\colon \mathbb{R}^S \to \mathbb{R}^N$ and $f\colon \mathbb{R}^N \to \mathbb{R}^M$ are differentiable functions. The *composite* function $f(g(\cdot))$ is also differentiable. Consider any point $x \in \mathbb{R}^S$. The chain rule allows us to evaluate the $M \times S$ derivative matrix of the composite function with respect to x, $D_x f(g(x))$ by matrix multiplication of the $N \times S$ derivative matrix of $g(\cdot)$, $Dg(x)$, and the $M \times N$ derivative matrix of $f(\cdot)$ evaluated at $g(x)$, that is, $Df(y)$, where $y = g(x)$. Specifically,

$$D_x f(g(x)) = Df(g(x))\, Dg(x). \tag{M.A.1}$$

The Product Rule

Here we simply provide a few illustrations.

(i) Suppose that $f\colon \mathbb{R}^N \to \mathbb{R}$ has the form $f(x) = g(x)h(x)$, where both $g(\cdot)$ and $h(\cdot)$ are real-valued functions of the N variables $x = (x_1, \ldots, x_N)$ (so that $g\colon \mathbb{R}^N \to \mathbb{R}$ and $h\colon \mathbb{R}^N \to \mathbb{R}$). Then the product rule of calculus tells us that

$$Df(x) = g(x)\, Dh(x) + h(x)\, Dg(x). \tag{M.A.2}$$

which, transposing, can also be written as

$$\nabla f(x) = g(x)\, \nabla h(x) + h(x)\, \nabla g(x).$$

(ii) Suppose that $f\colon \mathbb{R}^N \to \mathbb{R}$ has the form $f(x) = g(x)\cdot h(x)$ where both $g(\cdot)$ and $h(\cdot)$ are vector-valued functions which map the N variables $x = (x_1, \ldots, x_N)$ into $\mathbb{R}^M$. Then

$$Df(x) = g(x)\cdot Dh(x) + h(x)\cdot Dg(x). \tag{M.A.3}$$

Note that $h(x)\cdot Dg(x) = [h(x)]^{\mathrm{T}}\, Dg(x)$ is a $1 \times N$ matrix, as is the other term in the right-hand side. Thus, the vector-valued case (M.A.3) implies the scalar-valued formula (M.A.2).

(iii) Suppose that $f\colon \mathbb{R} \to \mathbb{R}^M$ has the form $f(x) = \alpha(x)g(x)$, where $\alpha(\cdot)$ is a real-valued function of one variable (i.e., $\alpha\colon \mathbb{R} \to \mathbb{R}$) and $g\colon \mathbb{R} \to \mathbb{R}^M$. Then

$$Df(x) = \alpha(x)\, Dg(x) + \alpha'(x)g(x). \tag{M.A.4}$$

(iv) Suppose that $f\colon \mathbb{R}^N \to \mathbb{R}^M$ has the form $f(x) = h(x)g(x)$ where $h\colon \mathbb{R}^N \to \mathbb{R}$ and $g\colon \mathbb{R}^N \to \mathbb{R}^M$. Then

$$Df(x) = h(x)\, Dg(x) + g(x)\, Dh(x). \tag{M.A.5}$$

Note that $g(x)$ is an M-element vector (i.e., an $M \times 1$ matrix) and $Dh(x)$ is a $1 \times N$ matrix. Hence, $g(x)\, Dh(x)$ is an $M \times N$ matrix (of rank 1). Observe also that (M.A.4) follows as a special case of (M.A.5).

M.B Homogeneous Functions and Euler's Formula

In this section, we consider functions of N variables, $f(x_1, \ldots, x_N)$, defined for all nonnegative values $(x_1, \ldots, x_N) \geq 0$.

Definition M.B.1: A function $f(x_1, \ldots, x_N)$ is *homogeneous of degree r* (for $r = \ldots, -1, 0, 1, \ldots$) if for every $t > 0$ we have

$$f(tx_1, \ldots, tx_N) = t^r f(x_1, \ldots, x_N).$$

As an example, $f(x_1, x_2) = x_1/x_2$ is homogeneous of degree zero and $f(x_1, x_2) = (x_1 x_2)^{1/2}$ is homogeneous of degree one.

Note that if $f(x_1, \ldots, x_N)$ is homogeneous of degree zero and we restrict the domain to have $x_1 > 0$ then, by taking $t = 1/x_1$, we can write the function $f(\cdot)$ as

$$f(1, x_2/x_1, \ldots, x_N/x_1) = f(x_1, \ldots, x_N).$$

Similarly, if the function is homogeneous of degree one then

$$f(1, x_2/x_1, \ldots, x_N/x_1) = (1/x_1) f(x_1, \ldots, x_N).$$

Theorem M.B.1: If $f(x_1, \ldots, x_N)$ is homogeneous of degree r (for $r = \ldots, -1, 0, 1, \ldots$), then for any $n = 1, \ldots, N$ the partial derivative function $\partial f(x_1, \ldots, x_N)/\partial x_n$ is homogeneous of degree $r - 1$.

Proof: Fix a $t > 0$. By the definition of homogeneity (Definition M.B.1) we have

$$f(tx_1, \ldots, tx_N) - t^r f(x_1, \ldots, x_N) = 0.$$

Differentiating this expression with respect to x_n gives

$$t \frac{\partial f(tx_1, \ldots, tx_N)}{\partial x_n} - t^r \frac{\partial f(x_1, \ldots, x_N)}{\partial x_n} = 0,$$

so that

$$\frac{\partial f(tx_1, \ldots, tx_N)}{\partial x_n} = t^{r-1} \frac{\partial f(x_1, \ldots, x_N)}{\partial x_n}.$$

By Definition M.B.1, we conclude that $\partial f(x_1, \ldots, x_N)/\partial x_n$ is homogeneous of degree $r - 1$. ■

For example, for the homogeneous of degree one function $f(x_1, x_2) = (x_1 x_2)^{1/2}$, we have $\partial f(x_1, x_2)/\partial x_1 = \frac{1}{2}(x_2/x_1)^{1/2}$, which is indeed homogeneous of degree zero in accordance with Theorem M.B.1.

Note that if $f(\cdot)$ is a homogeneous function of any degree then $f(x_1, \ldots, x_N) = f(x'_1, \ldots, x'_N)$ implies $f(tx_1, \ldots, tx_N) = f(tx'_1, \ldots, tx'_N)$ for any $t > 0$; that is, a radial expansion of a level set of $f(\cdot)$ gives a new level set of $f(\cdot)$.[1] This has an interesting implication: the slopes of the level sets of $f(\cdot)$ are unchanged along any ray through the origin. For example, suppose that $N = 2$. Then, assuming that $\partial f(\bar{x})/\partial x_2 \neq 0$, the slope of the level set containing point $\bar{x} = (\bar{x}_1, \bar{x}_2)$ at $\bar{x}$ is $-(\partial f(\bar{x})/\partial x_1)/(\partial f(\bar{x})/\partial x_2)$,

1. A *level set* of function $f(\cdot)$ is a set of the form $\{x \in \mathbb{R}^N_+ : f(x) = k\}$ for some k. A radial expansion of this set is the set of points obtained by multiplying each vector x in this level set by some positive scalar $t > 0$.

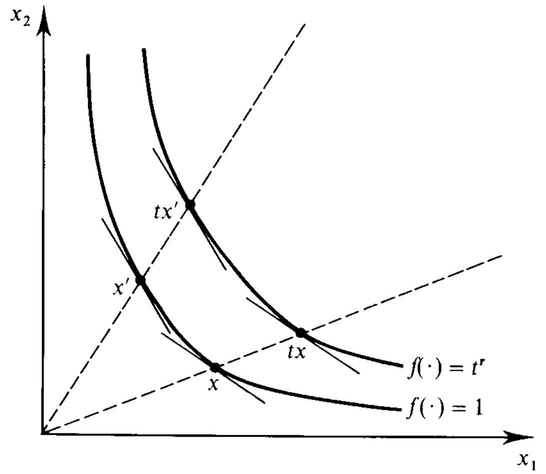


Figure M.B.1
The level sets of a homogeneous function.

and the slope of the level set containing point $t\bar{x}$ for $t > 0$ at $t\bar{x}$ is

$$-\frac{\partial f(t\bar{x})/\partial x_1}{\partial f(t\bar{x})/\partial x_2} = -\frac{t^{r-1}\,\partial f(\bar{x})/\partial x_1}{t^{r-1}\,\partial f(\bar{x})/\partial x_2} = -\frac{\partial f(\bar{x})/\partial x_1}{\partial f(\bar{x})/\partial x_2}.$$

An illustration of this fact is provided in Figure M.B.1.

Suppose that $f(\cdot)$ is homogeneous of some degree r and that $h(\cdot)$ is an increasing function of one variable. Then the function $h(f(x_1, \ldots, x_N))$ is called *homothetic.* Note that the family of level sets of $h(f(\cdot))$ coincides with the family of level sets of $f(\cdot)$. Therefore, for any homothetic function it is also true that the slopes of the level sets are unchanged along rays through the origin.

A key property of homogeneous functions is given in Theorem M.B.2.

Theorem M.B.2: (*Euler's Formula*) Suppose that $f(x_1, \ldots, x_N)$ is homogeneous of degree r (for some $r = \ldots, -1, 0, 1, \ldots$) and differentiable. Then at any $(\bar{x}_1, \ldots, \bar{x}_N)$ we have

$$\sum_{n=1}^{N} \frac{\partial f(\bar{x}_1, \ldots, \bar{x}_N)}{\partial x_n}\bar{x}_n = rf(\bar{x}_1, \ldots, \bar{x}_N),$$

or, in matrix notation, $\nabla f(\bar{x})\cdot\bar{x} = rf(\bar{x})$.

Proof: By definition we have

$$f(t\bar{x}_1, \ldots, t\bar{x}_N) - t^r f(\bar{x}_1, \ldots, \bar{x}_N) = 0.$$

Differentiating this expression with respect to t gives

$$\sum_{n=1}^{N} \frac{\partial f(t\bar{x}_1, \ldots, t\bar{x}_N)}{\partial x_n}\bar{x}_n - rt^{r-1}f(\bar{x}_1, \ldots, \bar{x}_N) = 0.$$

Evaluating at $t = 1$, we obtain Euler's formula. ■

For a function that is homogeneous of degree zero, Euler's formula says that

$$\sum_{n=1}^{N} \frac{\partial f(x_1, \ldots, x_N)}{\partial x_n}\bar{x}_n = 0.$$

As an example, note that for the function $f(x_1, x_2) = x_1/x_2$, we have $\partial f(\bar{x}_1, \bar{x}_2)/\partial x_1 = 1/\bar{x}_2$ and $\partial f(\bar{x}_1, \bar{x}_2)/\partial x_2 = -(\bar{x}_1/(\bar{x}_2)^2)$, and so

$$\sum_{n=1}^{N} \frac{\partial f(x_1, \ldots, x_N)}{\partial x_n} \bar{x}_n = \frac{1}{\bar{x}_2}\bar{x}_1 - \frac{\bar{x}_1}{(\bar{x}_2)^2}\bar{x}_2 = 0,$$

in accordance with Euler's formula.

For a function that is homogeneous of degree one, Euler's formula says that

$$\sum_{n=1}^{N} \frac{\partial f(x_1, \ldots, x_N)}{\partial x_n} \bar{x}_n = f(\bar{x}_1, \ldots, \bar{x}_N).$$

For example, when $f(x_1, x_2) = (x_1 x_2)^{1/2}$, we have $\partial f(\bar{x}_1, \bar{x}_2)/\partial x_1 = \frac{1}{2}(\bar{x}_2/\bar{x}_1)^{1/2}$ and $\partial f(\bar{x}_1, \bar{x}_2)/\partial x_2 = \frac{1}{2}(\bar{x}_1/\bar{x}_2)^{1/2}$, and so

$$\begin{aligned}\sum_{n=1}^{N} \frac{\partial f(x_1, \ldots, x_N)}{\partial x_n} \bar{x}_n &= \frac{1}{2}\left(\frac{\bar{x}_2}{\bar{x}_1}\right)^{1/2} \bar{x}_1 + \frac{1}{2}\left(\frac{\bar{x}_1}{\bar{x}_2}\right)^{1/2} \bar{x}_2 \\ &= (\bar{x}_1 \bar{x}_2)^{1/2} \\ &= f(\bar{x}_1, \bar{x}_2).\end{aligned}$$

M.C Concave and Quasiconcave Functions

In this section, we consider functions of N variables $f(x_1, \ldots, x_N)$ defined on a domain A that is a convex subset of $\mathbb{R}^N$ (such as $A = \mathbb{R}^N$ or $A = \mathbb{R}^N_+ = \{x \in \mathbb{R}^N : x \geq 0\}$).[2] We denote $x = (x_1, \ldots, x_N)$.

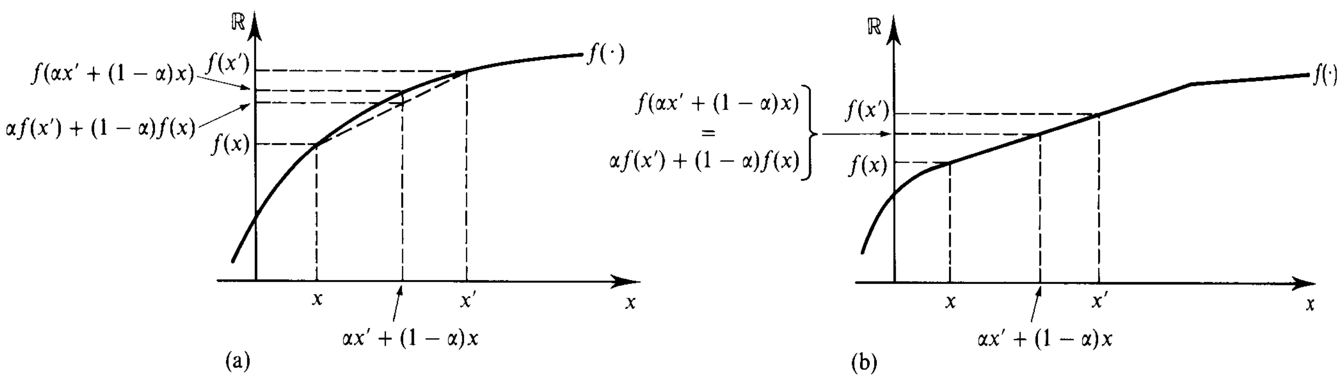
Definition M.C.1: The function $f: A \to \mathbb{R}$, defined on the convex set $A \subset \mathbb{R}^N$, is *concave* if

$$f(\alpha x' + (1 - \alpha)x) \geq \alpha f(x') + (1 - \alpha) f(x) \qquad \text{(M.C.1)}$$

for all x and $x' \in A$ and all $\alpha \in [0, 1]$. If the inequality is strict for all $x' \neq x$ and all $\alpha \in (0, 1)$, then we say that the function is *strictly concave*.

Figure M.C.1(a) illustrates a strictly concave function of one variable. For this case, condition (M.C.1) says that the straight line connecting any two points in the graph of $f(\cdot)$ lies entirely below this graph.[3] In Figure M.C.1(b), we show a function

Figure M.C.1 (a) A strictly concave function. (b) A concave but not strictly concave function.



2. For basic facts about convex sets, see Section M.G.
3. The *graph* of the function $f: A \to \mathbb{R}$ is the set $\{(x, y) \in A \times \mathbb{R}: y = f(x)\}$.

that is concave but not strictly concave; note that in this case the straight line connecting points x and x' lies on the graph of the function, so that condition (M.C.1) holds with equality.

We note that condition (M.C.1) is equivalent to the seemingly stronger property that

$$f(\alpha_1 x^1 + \cdots + \alpha_K x^K) \geq \alpha_1 f(x^1) + \cdots + \alpha_K f(x^K) \tag{M.C.2}$$

for any collection of vectors $x^1 \in A, \ldots, x^K \in A$ and numbers $\alpha_1 \geq 0, \ldots, \alpha_K \geq 0$ such that $\alpha_1 + \cdots + \alpha_K = 1$.

Let us consider again the one-variable case. We could view each number α_k in condition (M.C.2) as the "probability" that x^k occurs. Then condition (M.C.2) says that the value of the expectation is not smaller than the expected value. Indeed, a concave function $f: \mathbb{R} \to \mathbb{R}$ is characterized by the condition that

$$f\left(\int x \, \mathrm{d}F\right) \geq \int f(x) \, \mathrm{d}F \tag{M.C.3}$$

for any distribution function $F: \mathbb{R} \to [0, 1]$. Condition (M.C.3) is known as *Jensen's inequality.*

The properties of *convexity* and *strict convexity* for a function $f(\cdot)$ are defined analogously but with the inequality in (M.C.1) reversed. In particular, for a strictly convex function $f(\cdot)$, a straight line connecting any two points in its graph should lie entirely *above* its graph, as shown in Figure M.C.2. Note also that $f(\cdot)$ is concave if and only if $-f(\cdot)$ is convex.

Theorem M.C.1 provides a useful alternative characterization of concavity and strict concavity.

Theorem M.C.1: The (continuously differentiable) function $f: A \to \mathbb{R}$ is concave if and only if

$$f(x + z) \leq f(x) + \nabla f(x) \cdot z \tag{M.C.4}$$

for all $x \in A$ and $z \in \mathbb{R}^N$ (with $x + z \in A$). The function $f(\cdot)$ is strictly concave if inequality (M.C.4) holds strictly for all $x \in A$ and all $z \neq 0$.

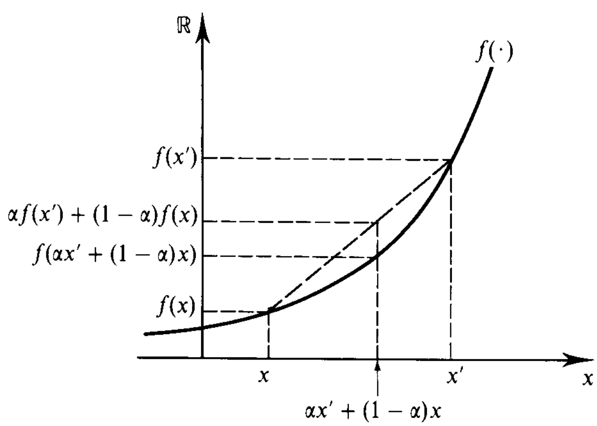


Figure M.C.2
A strictly convex function.

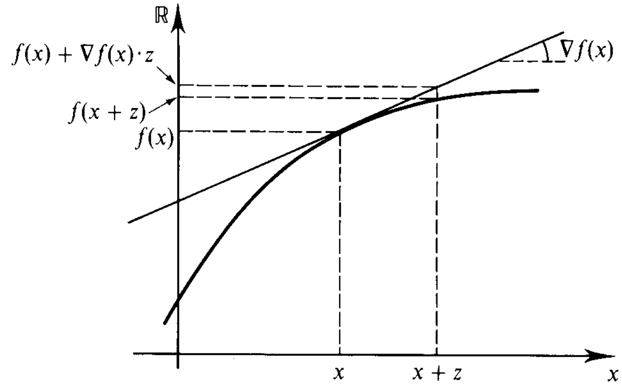


Figure M.C.3
Any tangent to the graph of a concave function lies above the graph of the function.

Proof: We argue only the necessity of condition (M.C.4) for concave functions. For all $\alpha \in (0, 1]$, the condition $f(\alpha x' + (1 - \alpha)x) \geq \alpha f(x') + (1 - \alpha)f(x)$ for all $x, x' \in A$, can be rewritten (think of $z = x' - x$) as

$$f(x + z) \leq f(x) + \frac{f(x + \alpha z) - f(x)}{\alpha}$$

for all $x \in A$, $z \in \mathbb{R}^N$ (with $x + z \in A$), and $\alpha \in (0, 1]$. Taking the limit as $\alpha \to 0$, we conclude that condition (M.C.4) must hold for a (continuously differentiable) concave function $f(\cdot)$. ■

Condition (M.C.4) is shown graphically in Figure M.C.3. It says that any tangent to the graph of a concave function $f(\cdot)$ must lie (weakly) above the graph of $f(\cdot)$.

The corresponding characterization of convex and strictly convex functions simply entails reversing the direction of the inequality in condition (M.C.4); that is, a convex function is characterized by the condition that $f(x + z) \geq f(x) + \nabla f(x) \cdot z$ for all $x \in A$ and $z \in \mathbb{R}^N$ (with $x + z \in A$).

We next develop a third characterization of concave and strictly concave functions.

Definition M.C.2: The $N \times N$ matrix M is *negative semidefinite* if

$$z \cdot Mz \leq 0 \tag{M.C.5}$$

for all $z \in \mathbb{R}^N$. If the inequality is strict for all $z \neq 0$, then the matrix M is *negative definite*. Reversing the inequalities in condition (M.C.5), we get the concepts of *positive semidefinite* and *positive definite* matrices.

We refer to Section M.E for further details on these properties of matrices. Here we put on record their intimate connection with the properties of the Hessian matrices $D^2 f(\cdot)$ of concave functions.[4]

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4. For theorems M.C.2, M.C.3, and M.C.4, the set A is assumed to be open (see Section M.F) so as to avoid boundary problems.

Theorem M.C.2: The (twice continuously differentiable) function $f: A \to \mathbb{R}$ is concave if and only if $D^2f(x)$ is negative semidefinite for every $x \in A$. If $D^2f(x)$ is negative definite for every $x \in A$, then the function is strictly concave.

Proof: We argue only necessity. Suppose that $f(\cdot)$ is concave. Consider a fixed $x \in A$ and a direction of displacement from x, $z \in \mathbb{R}^N$ with $z \neq 0$. Taking a Taylor expansion of the function $\phi(\alpha) = f(x + \alpha z)$, where $\alpha \in \mathbb{R}$, around the point $\alpha = 0$ gives

$$f(x + \alpha z) - f(x) - \nabla f(x) \cdot (\alpha z) = \frac{\alpha^2}{2} z \cdot D^2 f(x + \beta z) z$$

for some $\beta \in [0, \alpha]$. By Theorem M.C.1, the left-hand side of the above expression is nonpositive. Therefore, $z \cdot D^2 f(x + \beta z) z \leq 0$. Since α, hence β, can be taken to be arbitrarily small, this gives the conclusion $z \cdot D^2 f(x) z \leq 0$. ■

In the special case in which $N = 1$ [so $f(\cdot)$ is a function of a single variable], negative semidefiniteness of $D^2f(x)$ amounts to the condition that $d^2f(x)/dx^2 \leq 0$, whereas with negative definiteness we have $d^2f(x)/dx^2 < 0$ [to see this note that then $z \cdot D^2 f(x) z = z^2(d^2f(x)/dx^2)$]. Theorem M.C.2 tells us that in this case $f(\cdot)$ is concave if and only if $d^2f(x)/dx^2 \leq 0$ for all x, and that if $d^2f(x)/dx^2 < 0$ for all x, then $f(\cdot)$ is strictly concave. Note that Theorem M.C.2 does *not* assert that negative definiteness of $D^2f(x)$ must hold whenever $f(\cdot)$ is strictly concave. Indeed, this is not true: For example, when $N = 1$ the function $f(x) = -x^4$ is strictly concave, but $d^2f(0)/dx^2 = 0$.

For convex and strictly convex functions the analogous result to Theorem M.C.2 holds by merely replacing the word "negative" with "positive."

The remainder of this section is devoted to discussion of *quasiconcave* and *strictly quasiconcave* functions.

Definition M.C.3: The function $f: A \to \mathbb{R}$, defined on the convex set $A \subset \mathbb{R}^N$, is *quasiconcave* if its upper contour sets $\{x \in A: f(x) \geq t\}$ are convex sets; that is, if

$$f(x) \geq t \text{ and } f(x') \geq t \quad \text{implies that} \quad f(\alpha x + (1 - \alpha)x') \geq t \qquad \text{(M.C.6)}$$

for any $t \in \mathbb{R}$, $x, x' \in A$, and $\alpha \in [0, 1]$.[5] If the concluding inequality in (M.C.6) is strict whenever $x \neq x'$ and $\alpha \in (0, 1)$, then we say that $f(\cdot)$ is *strictly quasiconcave*.

Analogously, we say that the function $f(\cdot)$ is *quasiconvex* if its *lower* contour sets are *convex*; that is, if $f(x) \leq t$ and $f(x') \leq t$ implies that $f(\alpha x + (1 - \alpha)x') \leq t$ for any $t \in \mathbb{R}$, $x, x' \in A$, and $\alpha \in [0, 1]$. For strict quasiconvexity, the final inequality must hold strictly whenever $x \neq x'$ and $\alpha \in (0, 1)$. Note that $f(\cdot)$ is quasiconcave if and only if the function $-f(\cdot)$ is quasiconvex.

The level sets of a strictly quasiconcave function are illustrated in Figure M.C.4(a);

5. For more on convex sets, see Section M.G.

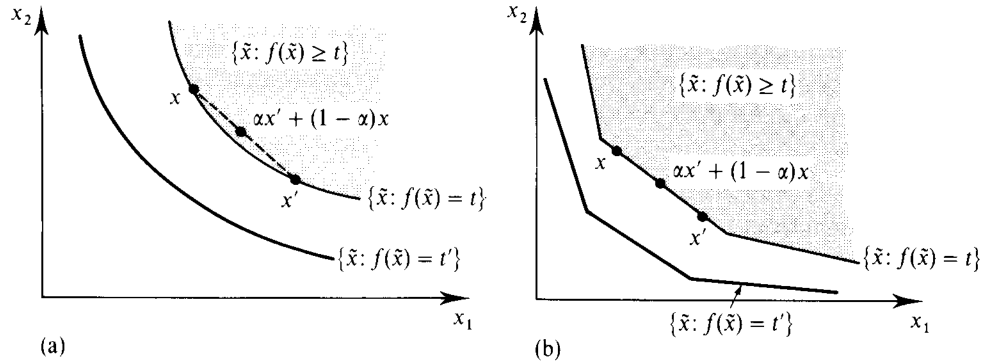

Figure M.C.4
(a) The level sets of a strictly quasiconcave function.
(b) The level sets of a quasiconcave function that is not strictly quasiconcave.

in Figure M.C.4(b) we show a function that is quasiconcave, but not strictly quasiconcave.

It follows from Definition M.C.3 that $f(\cdot)$ is quasiconcave if and only if

$$f(\alpha x + (1-\alpha)x') \geq \text{Min}\,\{f(x), f(x')\} \tag{M.C.7}$$

for all $x, x' \in A$ and $\alpha \in [0, 1]$. From this, or directly from (M.C.6), we see that a concave function is automatically quasiconcave. The converse is not true: For example, *any* increasing function of one variable is quasiconcave. Thus, concavity is a stronger property that quasiconcavity. It is also stronger in a different sense: concavity is a "cardinal" property in that it will *not* generally be preserved under an increasing transformation of $f(\cdot)$. Quasiconcavity, in contrast, will be preserved.

Theorems M.C.3 and M.C.4 are the quasiconcave counterparts of Theorems M.C.1 and M.C.2, respectively.

Theorem M.C.3: The (continuously differentiable) function $f\colon A \to \mathbb{R}$ is quasiconcave if and only if

$$\nabla f(x)\cdot(x' - x) \geq 0 \quad \text{whenever} \quad f(x') \geq f(x) \tag{M.C.8}$$

for *all* $x, x' \in A$. If $\nabla f(x)\cdot(x'-x) > 0$ whenever $f(x') \geq f(x)$ and $x' \neq x$, then $f(\cdot)$ is strictly quasiconcave. In the other direction, if $f(\cdot)$ is strictly quasiconcave and if $\nabla f(x) \neq 0$ for all $x \in A$, then $\nabla f(x)\cdot(x'-x) > 0$ whenever $f(x') \geq f(x)$ and $x' \neq x$.

Proof: Again, we argue only the necessity of (M.C.8) for quasiconcave functions. If $f(x') \geq f(x)$ and $\alpha \in (0, 1]$ then, using condition (M.C.7), we have that

$$\frac{f(\alpha(x'-x)+x) - f(x)}{\alpha} \geq 0.$$

Taking the limit as $\alpha \to 0$, we get $\nabla f(x)\cdot(x'-x) \geq 0$.

The need for the condition "$\nabla f(x) \neq 0$ for all $x \in A$" in the last part of the theorem is illustrated by the function $f(x) = x^3$ for $x \in \mathbb{R}$. This function is strictly quasiconcave (check this using the criterion in Definition M.C.3), but because $\nabla f(0) = 0$ we have $\nabla f(x)\cdot(x'-x) = 0$ whenever $x = 0$. ■

Theorem M.C.3's characterization of quasiconcave functions is illustrated in Figure M.C.5. The content of the theorem's condition (M.C.8) is that for any quasiconcave function $f(\cdot)$ and any pair of points x and x' with $f(x') \geq f(x)$, the gradient vector $\nabla f(x)$ and the vector $(x'-x)$ must form an acute angle.

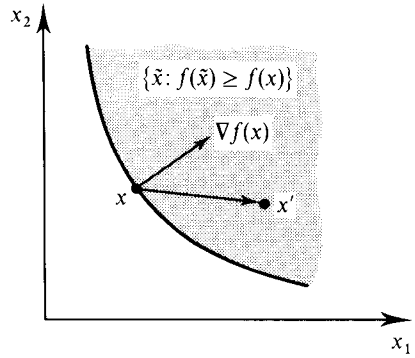

Figure M.C.5
Condition (M.C.8).

For a quasiconvex function, we reverse the direction of both inequalities in (M.C.8).

Theorem M.C.4: The (twice continuously differentiable) function $f: A \to \mathbb{R}$ is quasiconcave if and only if for every $x \in A$, the Hessian matrix $D^2f(x)$ is negative semidefinite in the subspace $\{z \in \mathbb{R}^N: \nabla f(x) \cdot z = 0\}$, that is, if and only if

$$z \cdot D^2f(x)z \leq 0 \quad \text{whenever} \quad \nabla f(x) \cdot z = 0 \tag{M.C.9}$$

for *every* $x \in A$.[6] If the Hessian matrix $D^2f(x)$ is negative definite in the subspace $\{z \in \mathbb{R}^N: \nabla f(x) \cdot z = 0\}$ for every $x \in A$, then $f(\cdot)$ is strictly quasiconcave.

Proof: Necessity (again, we limit ourselves to this) can be argued exactly as for Theorem M.C.2. The only adjustment is that we restrict z to be such that $\nabla f(x) \cdot z = 0$ and we resort to Theorem M.C.3 instead of Theorem M.C.1. ■

For a quasiconvex function, we replace the word "negative" with "positive" everywhere in the statement of Theorem M.C.4.

M.D Matrices: Negative (Semi)Definiteness and Other Properties

In this section, we gather various useful facts about matrices.

Definition M.D.1: The $N \times N$ matrix M is *negative semidefinite* if

$$z \cdot Mz \leq 0 \tag{M.D.1}$$

for all $z \in \mathbb{R}^N$. If the inequality is strict for all $z \neq 0$, then the matrix M is *negative definite*. Reversing the inequalities in condition (M.D.1), we get the concepts of *positive semidefinite* and *positive definite* matrices.

Note that a matrix M is positive semidefinite (respectively, positive definite) if and only if the matrix $-M$ is negative semidefinite (respectively, negative definite).

Recall that for an $N \times N$ matrix M the complex number λ is a *characteristic value* (or *eigenvalue* or *root*) if it solves the equation $|M - \lambda I| = 0$. The characteristic values of symmetric matrices are always real.

6. See Section M.E for a discussion of the properties of such matrices.

Theorem M.D.1: Suppose that M is an $N \times N$ matrix.

(i) The matrix M is negative definite if and only if the symmetric matrix $M + M^{\mathrm{T}}$ is negative definite.

(ii) If M is symmetric, then M is negative definite if and only if all of the characteristic values of M are negative.

(iii) The matrix M is negative definite if and only if M^{-1} is negative definite.

(iv) If the matrix M is negative definite, then for all diagonal $N \times N$ matrices K with positive diagonal entries the matrix KM is *stable*.[7]

Proof: Part (i) simply follows from the observation that $z \cdot (M + M^{\mathrm{T}})z = 2z \cdot Mz$ for every $z \in \mathbb{R}^N$.

The logic of part (ii) is the following. Any symmetric matrix M can be diagonalized in a simple manner: There is an $N \times N$ matrix of full rank C having $C^{\mathrm{T}} = C^{-1}$ and such that CMC^{T} is a diagonal matrix with the diagonal entries equal to the characteristic values of M. But then $z \cdot Mz = (Cz) \cdot CMC^{\mathrm{T}}(Cz)$, and for every $\hat{z} \in \mathbb{R}^N$ there is a z such that $\hat{z} = Cz$. Thus, the matrix M is negative definite if and only if the diagonal matrix CMC^{T} is. But it is straightforward to verify that a diagonal matrix is negative definite if and only if every one of its diagonal entries is negative.

Part (iii): Suppose that M^{-1} is negative definite and let $z \neq 0$. Then $z \cdot Mz = (z \cdot Mz)^{\mathrm{T}} = z \cdot M^{\mathrm{T}}z = (M^{\mathrm{T}}z) \cdot M^{-1}(M^{\mathrm{T}}z) < 0$.

Part (iv): It is known that a matrix A is stable if and only if there is a symmetric positive definite matrix E such that EA is negative definite. Thus, in our case, we can take $A = KM$ and $E = K^{-1}$. ■

For positive definite matrices, we can simply reverse the words "positive" and "negative" wherever they appear in Theorem M.D.1.

Our next result (Theorem M.D.2) provides a determinantal test for negative definiteness or negative semidefiniteness of a matrix M. Given any $T \times S$ matrix M, we denote by ${}_tM$ the $t \times S$ submatrix of M where only the first $t \leq T$ rows are retained. Analogously, we let M_s be the $T \times s$ submatrix of M where the first $s \leq S$ columns are retained, and we let ${}_tM_s$ be the $t \times s$ submatrix of M where only the first $t \leq T$ rows and $s \leq S$ columns are retained. Also, if M is an $N \times N$ matrix, then for any permutation π of the indices $\{1, \ldots, \mathrm{N}\}$ we denote by M^{π} the matrix in which rows and columns are correspondingly permuted.

Theorem M.D.2: Let M be an $N \times N$ matrix.

(i) Suppose that M is symmetric. Then M is negative definite if and only if $(-1)^r|{}_rM_r| > 0$ for every $r = 1, \ldots, N$.

(ii) Suppose that M is symmetric. Then M is negative semidefinite if and only if $(-1)^r|{}_rM_r^{\pi}| \geq 0$ for every $r = 1, \ldots, N$ *and* for every permutation π of the indices $\{1, \ldots, N\}$.

(iii) Suppose that M is negative definite (not necessarily symmetric). Then $(-1)^r|{}_rM_r^{\pi}| > 0$ for every $r = 1, \ldots, N$ and for every permutation π of the indices $\{1, \ldots, N\}$.[8]

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7. A matrix M is stable if all of its characteristic values have negative real parts. This terminology is motivated by the fact that in this case the solution of the system of differential equations $dx(t)/dt = Mx(t)$ will converge to zero as $t \to \infty$ for *any* initial position $x(0)$.

8. A matrix M such that $-M$ satisfies the condition in (iii) is called a P matrix. The reason is that the determinant of any submatrix obtained by deleting some rows (and corresponding columns) is positive.

Proof: (i) The necessity part is simple. Note that by the definition of negative definiteness we have that every ${}_rM_r$ is negative definite. Thus, by Theorem M.D.1, the characteristic values of ${}_rM_r$ are negative. The determinant of a square matrix is equal to the product of its characteristic values. Hence, $|{}_rM_r|$ has the sign of $(-1)^r$. The sufficiency part requires some computation, which we shall not carry out. It is very easy to verify for the case $N = 2$ [if the conclusion of (i) holds for a 2×2 symmetric matrix, then the determinant is positive and *both* diagonal entries are negative; the combination of these two facts is well known to imply the negativity of the two characteristic values].

For (ii), we simply note the requirement to consider all permutations. For example, if M is a matrix with all its entries equal to zero except the NN entry, which is positive, then M satisfies the nonnegative version of (i) but it is not negative semidefinite according to Definition M.D.1.

Notice that in part (iii) we only claim necessity of the determinantal condition. In fact, for nonsymmetric matrices the condition is not sufficient. ■

Example M.D.1: Consider a real-valued function of two variables, $f(x_1, x_2)$. In what follows, we let subscripts denote partial derivatives; for example, $f_{12}(x_1, x_2) = \partial^2 f(x_1, x_2)/\partial x_1\, \partial x_2$. Theorem M.C.2 tells us that $f(\cdot)$ is strictly concave if

$$D^2 f(x_1, x_2) = \begin{bmatrix} f_{11}(x_1, x_2) & f_{12}(x_1, x_2) \\ f_{21}(x_1, x_2) & f_{22}(x_1, x_2) \end{bmatrix}$$

is negative definite for all (x_1, x_2). According to Theorem M.D.2, this is true if and only if

$$|f_{11}(x_1, x_2)| < 0 \qquad \text{and} \qquad \begin{vmatrix} f_{11}(x_1, x_2) & f_{12}(x_1, x_2) \\ f_{21}(x_1, x_2) & f_{22}(x_1, x_2) \end{vmatrix} > 0,$$

or equivalently, if and only if

$$f_{11}(x_1, x_2) < 0$$

and

$$f_{11}(x_1, x_2) f_{22}(x_1, x_2) - [f_{12}(x_1, x_2)]^2 > 0.$$

Theorem M.C.2 also tells us that $f(\cdot)$ is concave if and only if $D^2 f(x_1, x_2)$ is negative semidefinite for all (x_1, x_2). Theorem M.D.2 tells us that this is the case if and only if

$$|f_{11}(x_1, x_2)| \le 0 \qquad \text{and} \qquad \begin{vmatrix} f_{11}(x_1, x_2) & f_{12}(x_1, x_2) \\ f_{21}(x_1, x_2) & f_{22}(x_1, x_2) \end{vmatrix} \ge 0,$$

and, permuting the rows and columns of $D^2 f(x_1, x_2)$,

$$|f_{22}(x_1, x_2)| \le 0 \qquad \text{and} \qquad \begin{vmatrix} f_{22}(x_1, x_2) & f_{21}(x_1, x_2) \\ f_{12}(x_1, x_2) & f_{11}(x_1, x_2) \end{vmatrix} \ge 0.$$

Thus, $f(\cdot)$ is concave if and only if

$$f_{11}(x_1, x_2) \le 0,$$

$$f_{22}(x_1, x_2) \le 0,$$

and

$$f_{11}(x_1, x_2) f_{22}(x_1, x_2) - [f_{12}(x_1, x_2)]^2 \ge 0. \qquad ■$$